

SEOSAW dataset manual

John L. Godlee

21st January 2021

Contents

1	Introduction	2
2	SEOSAW code repository	2
3	Description of the SEOSAW dataset	3
3.1	Descriptive statistics and dataset visualisation	3
3.2	Plot data	9
3.2.1	Nesting and subplots	9
3.2.2	Stems or trees	11
3.2.3	Permanent or one-off plot	11
3.2.4	Sampling metadata	12
3.2.5	Land use metadata	12
3.2.6	Fire and herbivory metadata	13
3.3	Stem data	13
3.3.1	Taxonomy	13
3.3.2	Biomass	15
3.3.3	Diversity	15
3.4	Spatial data	15
3.4.1	Plot polygons	15
3.4.2	Spatial data extraction	15
3.5	Dataset caveats	17
4	Contributing data	18
4.1	Submission guidelines	18
4.2	Vital data	18
4.3	Additional data	19

5	Cleaning data for the SEOSAW dataset	19
5.1	Cleaning plot data	22
5.1.1	Rename columns, replace values	22
5.1.2	Check column contents are logically consistent	22
5.1.3	Plot polygons	24
5.2	Cleaning stem data	24
5.2.1	Rename columns, replace values	24
5.2.2	Check column contents are logically consistent	25
5.2.3	Fix species names	26
5.2.4	Generate new columns	27
5.2.5	Check logical consistency of plots and stems tables	29
5.2.6	Generate plot level statistics from stem data	29
5.3	Compiling datasets	30
6	Dataset table columns	32

1 Introduction

This manual focuses on the **SEOSAW** dataset and provides information on three main topics:

1. The contents of the SEOSAW dataset
2. Contributing data to SEOSAW
3. How data is cleaned for the SEOSAW dataset with the `{seosawr}` R package

For further guidance on collecting SEOSAW data in the field, you should consult the **SEOSAW Field Manual**.

2 SEOSAW code repository

The **SEOSAW code repository** is currently hosted on BitBucket. It contains the `{seosawr}` R package, reproducible code used to clean data from each site, and code for further SEOSAW analyses. Below is a breakdown of directories in the repository and what they contain:

- **cleaning** - R scripts which clean data for the SEOSAW dataset
- **analysis** - directories containing analyses conducted by SEOSAW researchers
- **seosawr** - the SEOSAW R package, used to clean the SEOSAW data and provides convenience functions for further analysis

- `doc` - documents, manuals, and items for the [SEOSAW website](#)

Note that the SEOSAW dataset lives in a separate repository with limited private access, due to the sensitivity of some datasets.

To access the SEOSAW code repository, use `git clone` or download the repository directly from BitBucket as a `.zip` file.

3 Description of the SEOSAW dataset

This section gives a brief overview of the contents of the latest iteration of the SEOSAW dataset, which is available for research purposes upon approval from the SEOSAW network. At the time of writing the latest version is `v2.10`.

The SEOSAW dataset is built around two key tables:

- `plots.csv` - Plot level data
- `stems.csv` - Stem level data

Each row in the plots table represents a single plot. Each row in the stems table represents a single stem measurement. Note that a single tree may contain many stems and each stem may be measured multiple times. See [section 6](#) for a breakdown of columns found in the stems table and plots table. All data objects are linked by the `plot_id` column, which serves as the primary key for the dataset.

Auxiliary data objects in the SEOSAW data repository:

- `polys.shp` - Polygons for each plot
- `points.shp` - Point locations for each plot
- `census.csv` - Plot level data calculated for each census
- `stems_latest.csv` - Stem data subset for the most recent census of each plot
- `stems_multi.csv` - Stem data subset for plots with multiple censuses
- `plots_spatial.csv` - Data extracted from various spatial data layers for each plot
- `species_plot_matrix.csv` - Matrix of stem abundance by species and plot

3.1 Descriptive statistics and dataset visualisation

- Number of plots: 10015
- Number of permanent plots: 424
- Number of one-off plots: 9591
- Number of stem measurements: 538437
- Number of species: 1139
- Total plot area surveyed (km^2): 13.6

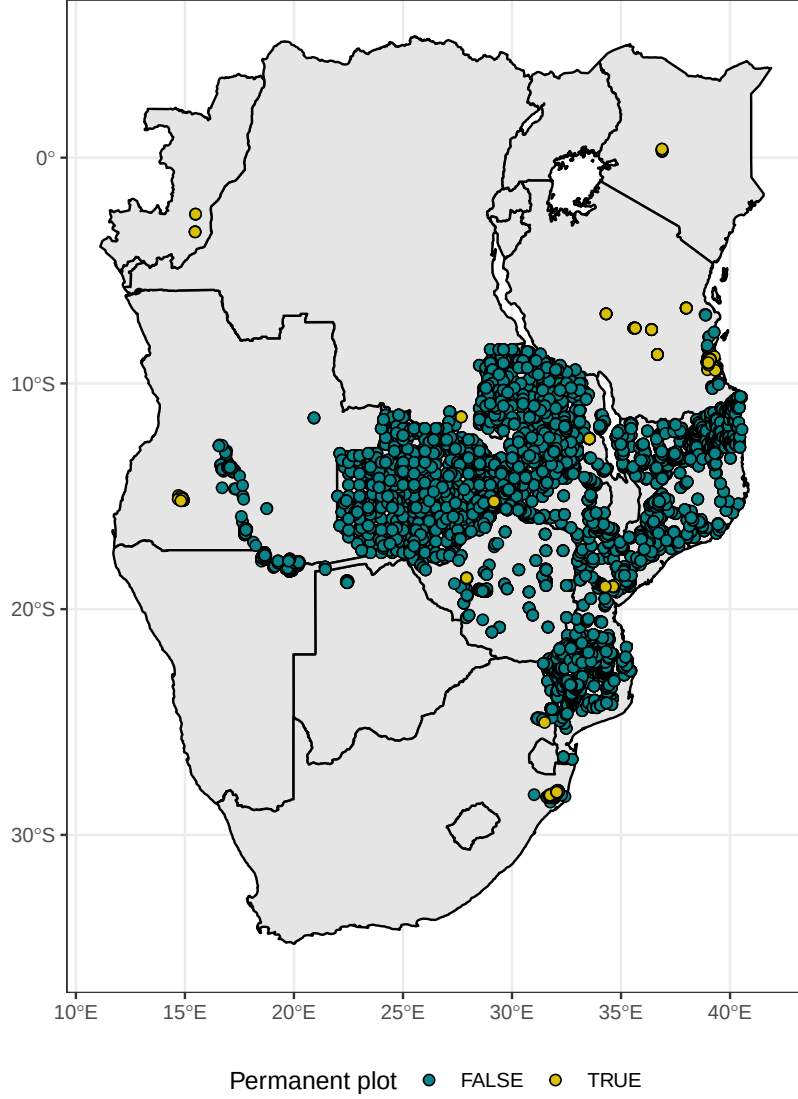


Figure 1: Distribution of plots in southern Africa.

Table 1: Country level statistics. Note that none of these statistics are representative of that country, they are only representative of the plots within those countries, which are themselves not evenly distributed.

Country	Mean AGB ($tC\ ha^{-1}$)	Mean BA ($m^2\ ha^{-1}$)	N plots
Angola	56.168	10.6322	332
Congo - Brazzaville	5.448	0.7744	6
Congo - Kinshasa	110.582	18.1995	15
Kenya	18.175	10.3406	6
Malawi	56.319	8.4628	181
Mozambique	63.916	10.3158	4030
Namibia	93.519	22.0113	213
South Africa	318.936	53.2357	586
Tanzania	64.265	10.2160	132
Zambia	63.187	9.7989	4113
Zimbabwe	90.384	19.0349	361

Table 2: Dataset level statistics.

Dataset ID	Country	N plots	Min. DBH threshold (cm)	Plot shape	Plot area (ha)	Mean AGB ($tC\ ha^{-1}$)	Mean BA ($m^2\ ha^{-1}$)
ABG	Angola	15	5	rectangle	1	48.0	8.9
ACM	Angola	60	1	rectangle	0.1	83.8	12.4
CBN	Congo - Brazzaville	6	9, 11	rectangle	25	5.4	0.8
CLP	Congo - Kinshasa	12	5	circle	0.03	111.8	18.8
DKS	Congo - Kinshasa	3	10	rectangle	10, 1	105.9	15.7
KLS	Kenya	6	0	rectangle	0.25	18.2	10.3
MAR	Mozambique	384	5	circle	0.13, 0.07, 0.03, 0.2	29.9	6.5
MCF	Malawi	93	20	circle	0.1, NA, 0.13, 0.11	31.6	5.4
MCL	Malawi	38	5	rectangle	0.25	56.4	10.5
MGR	Mozambique	15	5	rectangle	1	55.2	9.2
MLC	Mozambique	51	5	circle	0.03	124.3	19.0
MMB	Mozambique	10	5	rectangle	1	115.0	15.0
MNF	Malawi	50	20	circle	0.11, 0.1, 0.12, 0.13	101.9	12.4
MNI	Mozambique	3520	10	rectangle	0.1	67.6	10.7
MNR	Mozambique	50	5	circle	0.28	18.9	2.9
NCC	Namibia, Angola	217	NA, 20, 45	circle	0.03, 0.07	107.6	24.8
SBT	South Africa	55	4	circle	0.1	7.2	4.3
SHD	South Africa	175	6	rectangle	0.25	746.3	127.7
SHS	South Africa	202	10	rectangle	0.04	177.8	23.6
SSM	South Africa	1	NA	rectangle	4	5.3	1.1
SST	South Africa	153	NA	circle	0.01	5.9	2.7
TIM	Tanzania	45	5	rectangle	0.04	76.7	13.6
TKM	Tanzania	12	5	rectangle	0.09	60.9	10.6
TKW	Tanzania	25	5	rectangle	1	55.6	7.8
TKX	Tanzania	14	40	rectangle	9	22.1	2.2
TLA	Tanzania	36	5	rectangle	0.5, 0.55, 0.65, 1.5, 1.6, 0.9, 1, 0.75, 1.9, 0.7, 0.2, 0.3, 0.4, 0.1, 0.8, 1.1, 0.6	72.3	10.7
VAS	Angola, Namibia, Botswana	293	NA, 19, 5, 10	rectangle	0.1	34.6	7.3
ZCC	Zambia	16	1	rectangle	0.02	67.5	12.7
ZGF	Zimbabwe	71	5	circle	0.02	225.3	43.1
ZHF	Zimbabwe	7	5	circle	0.02	162.7	38.5
ZHH	Zimbabwe	18	5	rectangle	0.1	19.5	13.6
ZIS	Zambia	3886	10	rectangle	0.1	46.3	7.7
ZKS	Zambia	33	5	circle	0.04, 0.11, 0.03, 0.1, 0.06, 0.05, 0.08, 0.12, 0.09, 0.02, 0.13, 0.07	27.9	7.1
ZLT	Zambia	178	10	circle	0.02	388.4	48.8
ZNF	Zimbabwe	135	10	rectangle	0.1	26.0	5.2
ZPF	Zimbabwe	15	5	rectangle	0.25	147.4	24.2
ZRF	Zimbabwe	35	5	circle	0.01	201.4	44.7
ZWP	Zimbabwe	80	3	circle	0.13	20.9	5.4

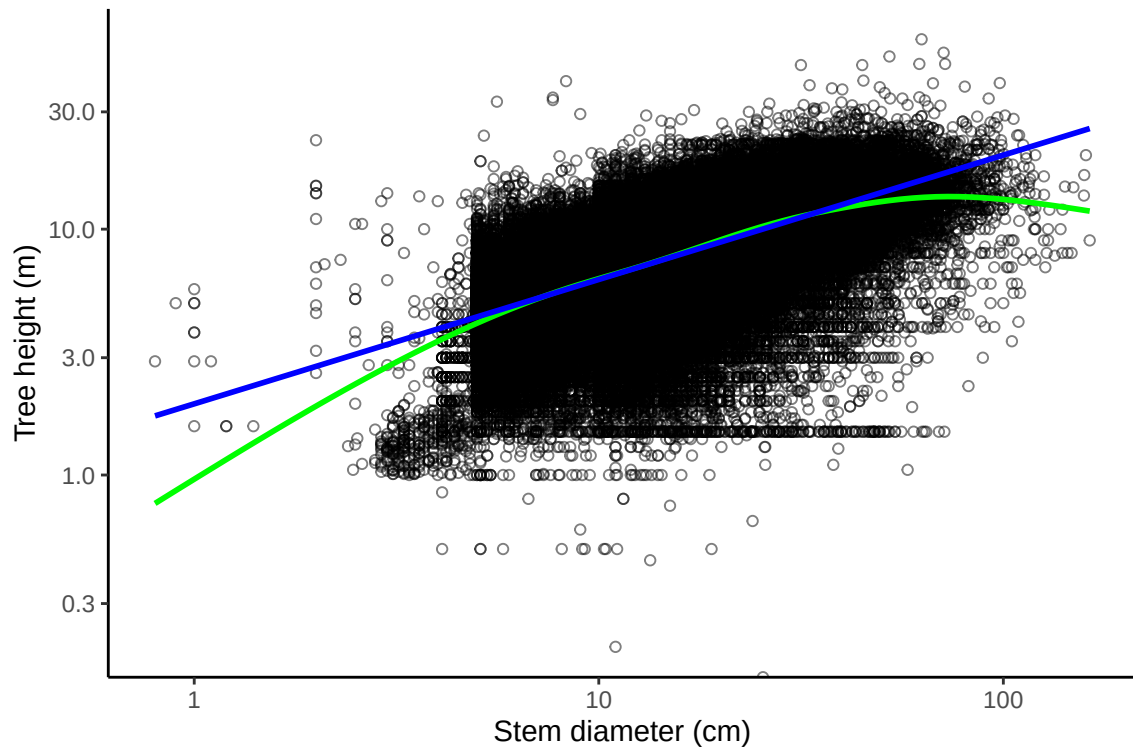


Figure 2: Diameter-height distribution of all stem measurements. The blue line shows a linear regression and the green line shows a loess line of best fit.

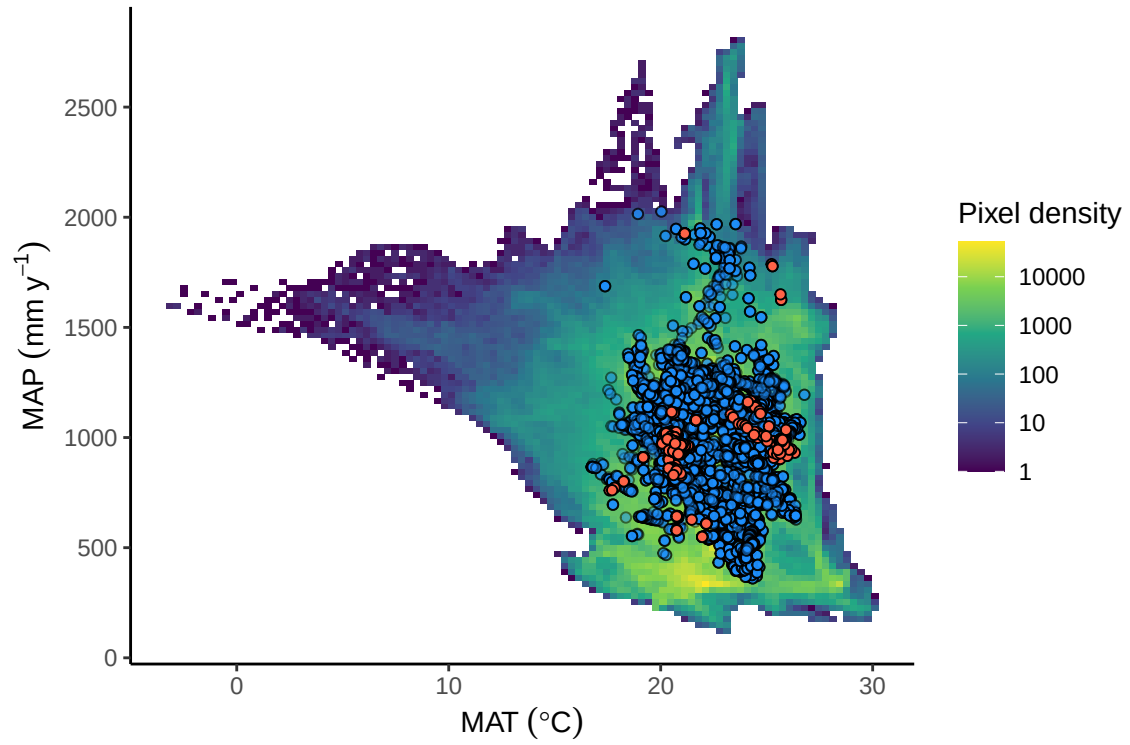


Figure 3: Distribution of plots according to Mean Annual Temperature (MAT) and Mean Annual Precipitation (MAP) of the SEOSAW region. MAT and MAP extracted from the WorldClim dataset gridded at 30 arc second resolution, between 1970 and 2000 (Fick and Hijmans, 2017). Permanent plots are coloured red, non-permanent plots are blue.

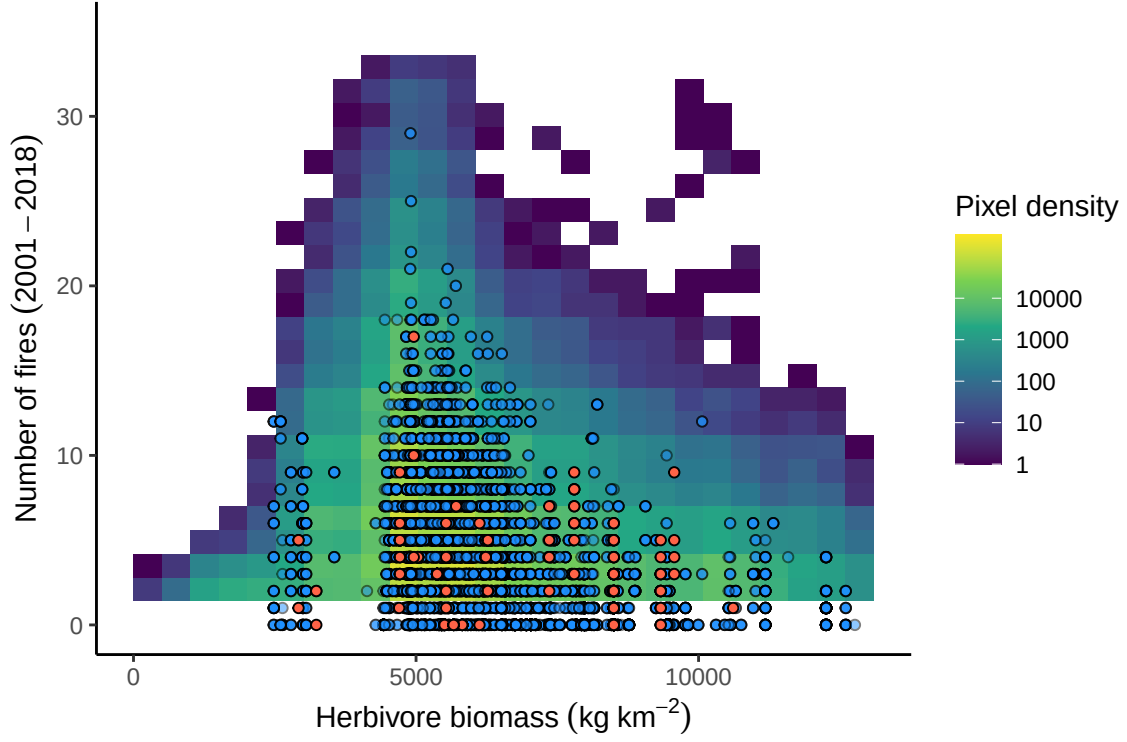


Figure 4: Distribution of plots according to herbivore biomass and fire frequency within the SEOSAW region. Herbivore biomass is derived from Hempson, Archibald and Bond (2017) and estimates the biomass of all herbivores. Fire count is derived from the MODIS Burned Area Product (MCD64A1) between 2001 and 2018 (L. Giglio et al., 2015). Both products are gridded at 0.05 degree resolution. Permanent plots are coloured red, non-permanent plots are blue.

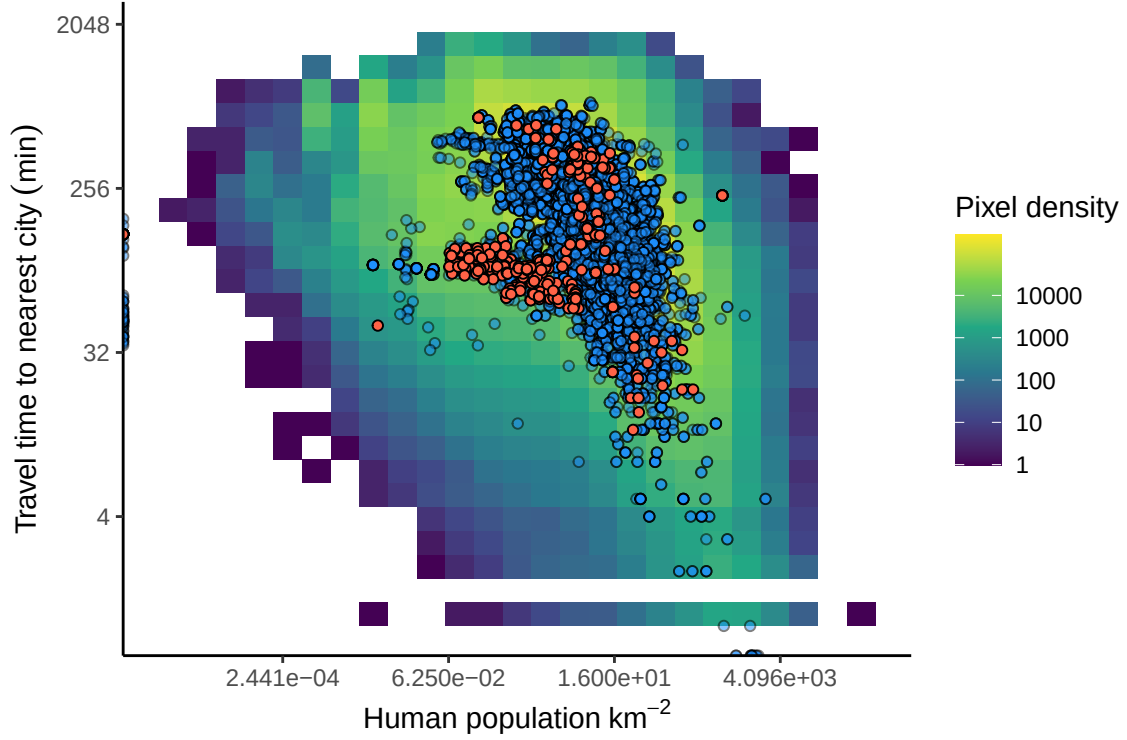


Figure 5: Distribution of plots according to human population density and travel time to cities within the SEOSAW region. Human population density data is taken from the WorldPop dataset (1 km) (Linard et al., 2012). Travel time data is estimated from the Malaria Atlas Project 2015 travel time to cities product (Meijer et al., 2018) (30sec). Note that both axes are log-transformed. Permanent plots are coloured red, non-permanent plots are blue.

3.2 Plot data

This section describes important aspects of the plot data in `plots.csv`. See [section 6](#) for a full breakdown of columns in the plots table.

3.2.1 Nesting and subplots

It is common for researchers to stratify their sampling effort across a plot, often according to stem diameter classes, to reduce the work required to measure a plot. This stratification is often referred to as ‘nesting’. See the diagram below for an example of a nested plot. In this example, there are three levels of nesting. Across the whole plot (0.2 ha) stems >10 cm DBH are measured. Within the first level of nesting (N1, 0.1 ha), stems >5 cm DBH are measured, within the second level (N2, 0.05 ha) stems >2 cm DBH, and within the third level (N3, 0.025 ha) stems >1 cm DBH.

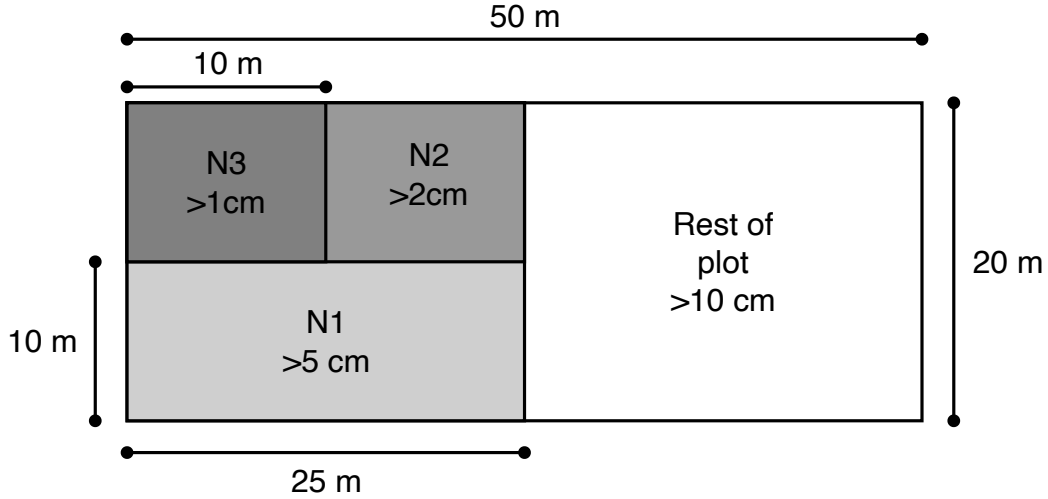


Figure 6: Schematic diagram of a nested plot with three levels of nesting.

There are a number of columns in `plots.csv` which record nesting. The columns listed below are only those to describe the first level of nesting, but there are similar columns for the second (`n2_`) and third (`n3_`) levels of nesting:

- `nested_1` - is the plot nested at the first level?
- `n1_min_diam_thresh` - minimum stem diameter threshold within level of nesting
- `n1_min_height_thresh` - minimum stem height threshold within level of nesting
- `n1_plot_shape` - shape of nested plot
- `n1_plot_width` - width of nested plot
- `n1_plot_length` - length of nested plot
- `n1_plot_diameter` - diameter of nested plot

If plots are nested, with variation in sampling effort for trees of different sizes, the Fractional Probability of Inclusion in the `fpc` column of the stem table describes the sampling effort for each stem. In the example, stems >10 cm DBH are measured across the whole plot, so their `fpc` is 1.0. Stems >5 cm DBH are measured in an area 20×25 m (0.1 ha), which is half the total plot area, so their `fpc` is 0.5. Stems >2 cm are measured over 0.05 ha, which is a quarter of the total plot area, so their `fpc` is 0.25, and so on. At present, the SEOSAW dataset only allows up to three levels of nesting, but this may change in the future.

It is important to distinguish between nesting and subplots. Nested plots, as demonstrated above, are smaller parcels within a plot which denote variation in sampling effort. In contrast, subplots are merely administrative sub-divisions of a plot which can help with relocating stems for future censuses, for example. Sampling effort should be the same across all subplots. Below is an example of a plot with four 20×25 m subplots.

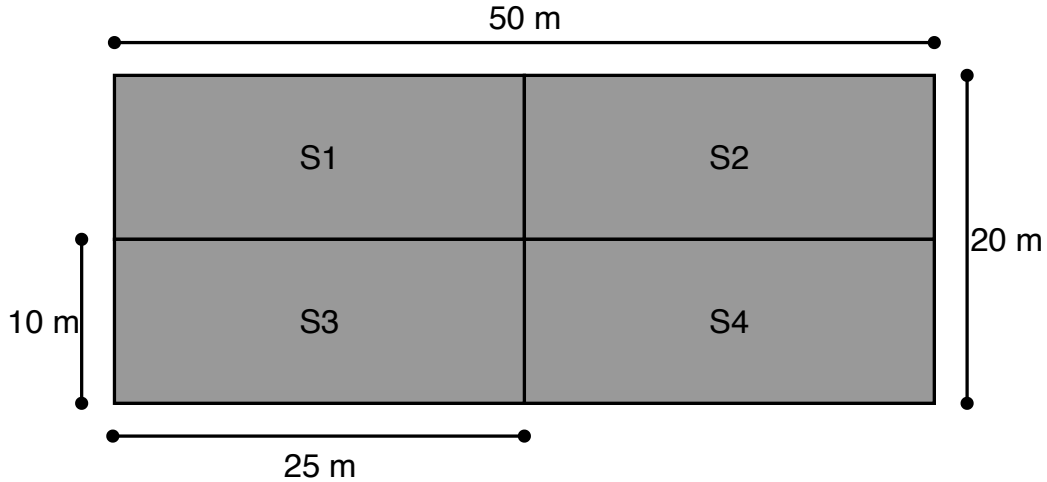


Figure 7: Schematic diagram of a plot with subplots, each occupying 25% of the total plot space.

There are a few columns in `plots.csv` which record subplot dimensions. It is assumed that all subplots have the same dimensions and that subplots are not nested:

- `subplot_n` - number of subplots in the plot
- `subplot_shape` - shape of subplots
- `subplot_length` - length of subplots
- `subplot_width` - width of subplots
- `subplot_diameter` - diameter of subplots

3.2.2 Stems or trees

The ideal data for the SEOSAW dataset is organised so that there is one measurement for each stem, and stems are grouped into trees with a permanent tag to identify each stem. We understand however that not all data matches this ideal, and is dependent on the original data collection protocol. There are a few columns in the `plots` table to help describe how stem data is organised:

- `tags` - Do stems have permanent tags which allow for remeasurement?
- `tree_stem` - Are measurements in the stem table recorded at the tree level or stem level?
- `tree_diff` - If `tree_stem == 'stem'`, were stem measurements grouped into separate trees?

3.2.3 Permanent or one-off plot

A permanent plot in the SEOSAW dataset is defined as one where trees and new recruits can be measured at a later date. The boundaries of a permanent plot must be clearly identifiable, i.e. with corner posts for a rectangular plot or a central marker for a circular plot. Individual trees must also be identifiable in a permanent plot. Normally this is achieved through the use of numbered tags, but can alternatively be achieved with a highly accurate grid coordinate system, a differential GPS, or other tagging methods.

One-off plots are those where only a single census has been made, with no way of exactly identifying plot boundaries or tree identity in the future.

Both permanent and one-off plots are accepted by SEOSAW. The `permanent` column in the plots table notes whether a plot is permanent or not.

3.2.4 Sampling metadata

There is variation in measurement protocol across the SEOSAW dataset. In some plots for example, only living trees were sampled while dead trees were ignored completely. This does not mean there were no dead trees in those plots. Unless these differences in measurement protocol are accounted for in analysis, erroneous relationships may appear. There are a few columns in the plots table which describe variations in measurement protocol:

- `lianas_sampled`
- `soil_sampled`
- `understory_sampled`
- `dead_stems_sampled`
- `stumps_sampled`
- `sampling_notes`
- `sampling_design_code` - Reference to document containing the description of protocol for data collection.

3.2.5 Land use metadata

Vegetation within a plot may differ hugely based on any human land use. The following columns in the plots table provide metadata on human land use and may be used to filter data prior to analysis:

- `pa` - protected area
- `pa_name`
- `pa_type`
- `wdpa_id` - WDPA protected area ID
- `local_community_rights`
- `other_user_rights`
- `timber_harvesting`
- `high_graded_100_years`
- `fuel_wood_harvesting`
- `charcoal_harvesting`
- `other_woody_product_harvesting`
- `ntfp_harvesting`
- `farmed_30_years`
- `land_use_notes`

3.2.6 Fire and herbivory metadata

Disturbance by fire and herbivory can greatly affect vegetation dynamics. Users of the SEOSAW dataset may want to filter plots based on disturbance regime. These columns in the plots table provide metadata on disturbances:

- `fire_exclusion` - Are fires excluded by humans?
- `fire_exclusion_success`
- `fire_treatment` - Are fires controlled by humans?
- `fire_return_interval`
- `fire_treatment_season`
- `fire_treatment_notes`
- `elephant_exclosure` - Are elephants excluded by humans?
- `elephants`
- `elephant_density`
- `elephant_data_source`
- `exclosure`
- `small_nonsocial_browsers`
- `large_browsers`
- `medium_sized_social_mixed_diet`
- `nonruminants_excluding_suids`
- `water_dependent_grazers`
- `goat_grazing`
- `cattle_grazing`
- `herbivory_notes`

3.3 Stem data

This section describes aspects of the stem data in `stems.csv`. See [section 6](#) for a full breakdown of columns in the stems table.

3.3.1 Taxonomy

There are a number of columns in the stems table which describe the species of a stem:

- `species_orig_binom` - binomial species name supplied by data originator
- `species_orig_local` - local (non-binomial) species name supplied by data originator
- `genus` - genus portion of binomial name supplied by data originator
- `species` - species epithet of binomial name supplied by data originator

- `confer` - “cf. ...” portion of binomial name supplied by data originator
- `subspecies` - “subsp. ...” portion of binomial name supplied by data originator
- `variety` - “var. ...” portion of binomial name supplied by data originator
- `species_name_clean` - combined binomial species name after cleaning
- `family_clean` - family name after cleaning
- `genus_clean` - genus name after cleaning
- `species_clean` - species name after cleaning
- `cjb_name` - Binomial species name matched by the CJB African plants database
- `cjb_url` - URL of entry matched by the CJB African plants database

The taxonomic ‘backbone’ of the SEOSAW dataset is the [CJB African plants database](#) with some notable exceptions, which are discussed below. The dataset also uses the `taxize::gnr_resolve()` utility to query multiple databases to check species names and provide higher taxonomic rank information. The queried databases are:

- [Open Tree of Life Reference Taxonomy](#)
- [The International Plant Names Index](#)
- [Tropicos - Missouri Botanical Garden](#)
- [GBIF Backbone Taxonomy](#)
- [NCBI](#)

The breakup of the *Acacia* genus has been well documented by (Kyalangalilwa et al., 2013). SEOSAW uses their list of synonyms to correct ex-*Acacia* species. The CJB database does not currently acknowledge the *Acacia* split, so queries to CJB retain the *Acacia* name.

All *Rhus* species in SEOSAW’s study region belong to *Searsia* and are transferred in this way. All recent phylogenetic results point to this, but the formal taxonomic work to transfer African *Rhus* species is incomplete.

Previously, *Marquesia* species were placed in the *Trillesanthus* genus, however this has since been reverted. SEOSAW corrects *Trillesanthus* to *Marquesia*.

The latest taxonomy states that *Maytenus* is a strictly Neotropical genus, while acknowledging that African *Maytenus* are a “problem”. The latest taxonomic literature doesn’t tell us what to do with African *Maytenus*. SEOSAW defaults to the CJB taxonomy on *Gymnosporia*/*Maytenus*, but this may change in the future.

There are also some miscellaneous synonym corrections which are maintained throughout the canonical SEOSAW dataset. The vast majority of these corrections reflect previously debunked names which still appear in older datasets.

If a species cannot be identified, it is given the “indet” epithet. If only the genus is known, it is recorded as “*Genus* indet” (e.g. *Brachystegia* indet). If only the family is known, it is recorded as “Family indet” (e.g. Fabaceae indet). As a last resort, if only the local name of the species is known, it is recorded in the `species_orig_local` column.

3.3.2 Biomass

Biomass, specifically aboveground woody biomass (AGB) is stored in the `agb` column in the stems table. Additionally, plot level biomass is stored in the `agb` and `agb_ha` (AGB per hectare) columns in the plots table. It is calculated in the SEOSAW dataset by default using an allometric equation devised by (Chave, Réjou-Méchain et al., 2014) which takes into account species wood density, stem diameter and latitude/longitude.

3.3.3 Diversity

The `{seosawr}` package provides slim wrappers around existing functions in the `{vegan}` package to calculate biodiversity metrics at the plot level:

- `richness` - Species richness
- `rarefy` - Rarefied species richness - `vegan::rarefy()`
- `shannon` - Shannon-Wiener index - $H' = -\sum_{i=1}^s p_i \ln p_i$
- `simpson` - Simpson's index - $D = \sum_{i=1}^s p_i^2$
- `invsimpson` - Inverse Simpson's index - $1/D$
- `evenness` - Simpson's evenness - $(1/D)/N$

Additionally, each plot's five most dominant species, estimated simply by tree abundance, are shown in `dom_1` etc., with their proportional abundances in `prop_1` etc. The values returned are the proportional abundance of the corresponding species, within the plot.

3.4 Spatial data

3.4.1 Plot polygons

Plot outline polygons are provided in `polys.shp`, for each plot where appropriate location and plot dimensions data is available. For rectangular plots, these polygons are normally created from GPS coordinates taken at each corner and used as vertices when transforming to a closed spatial polygon. Plot point information is stored in `points.shp`, and uses the plot centre location. Plot polygons are automatically generated using the `polyGen()` function in the `{seosawr}` R package, which uses the plot centre coordinates, plot shape, and plot dimensions to create an approximate polygon if none are provided.

3.4.2 Spatial data extraction

`plots_spatial.csv` provides data extracted from a number of spatial data layers for each plot. See [section 6](#) for a breakdown of these spatial data layers and how they are used.

The SEOSAW region is provided as a spatial data object in the `{seosawr}` R package (`data(seosaw_region)`), along with a simplified bounding box of the SEOSAW region (`data(seosaw_bbox)`). There is also a useful outline of all countries in Africa (`data(africa)`).

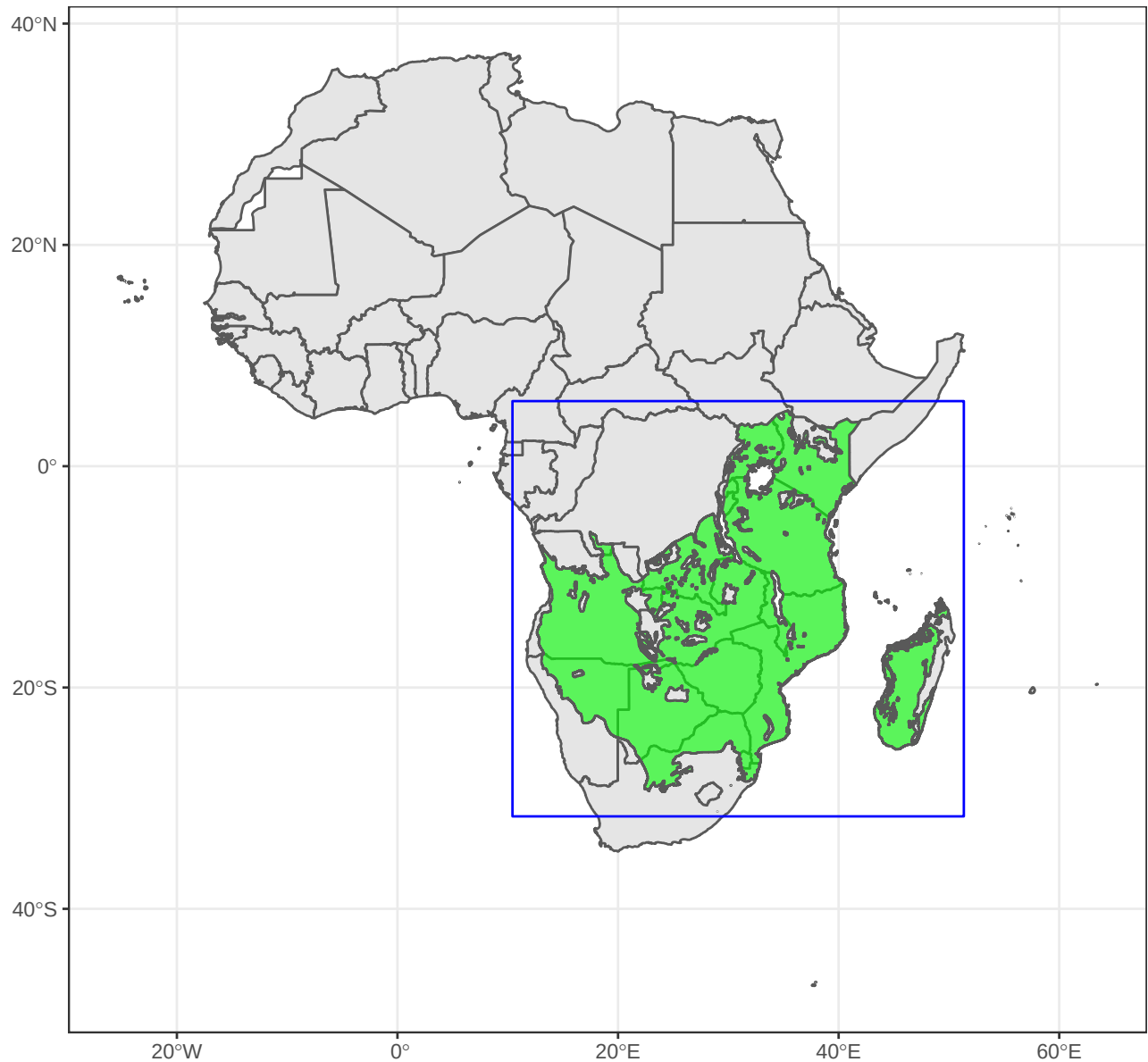


Figure 8: The spatial extent of the SEOSAW region shown in green and the SEOSAW bounding box as a blue rectangle, within Africa.

The SEOSAW region is based largely on White's vegetation map, which defines the locations of savannas and woodlands (White, 1983).

A key issue was defining the north-eastern boundary of the SEOSAW region. It is defined here as the borders of Kenya and Uganda, excluding Ethiopia, Somalia, and South Sudan. While the decision to use country boundaries was largely pragmatic, it is backed up by environmental data. The mountainous Ethiopian highlands on the border between Kenya, Ethiopia and Somalia cause quite a clear change in precipitation, probably due to them blocking dry Arabian air moving further southwest.

The north-western boundary largely follows the rift valley lakes, with everything to the west of them being excluded. Some areas to the east of the lakes are also excluded as they are classed by White's vegetation map as

Congo Forest Transitions and Mosaics.

The south-east of the region is further reduced from White’s vegetation map according to the Terrestrial Ecoregions of the World (Dinerstein et al., 2017). White classes some areas as woodland or forest in this area, but for our purposes they are considered too temperate, and are thus excluded from the SEOSAW region.

3.5 Dataset caveats

The SEOSAW dataset is aggregated from many individual datasets contributed by researchers around the world. While effort has been made to ensure that datasets are compatible with each other, particularly with respect to stem data like `stem_diam` and `height`, some pieces of metadata should not be compared between datasets. For example, `canopy_cover` is collected with hemispherical photos, different types of densitometer, aerial photos and many other methods. Due to the subjectivity of all these methods, it is not recommended to use this data as a variable in statistical analysis when considering multiple datasets; it should be used only as metadata. Below is a non-exhaustive list of fields we recommend not to compare between sites in analyses:

- `slope` - Variable accuracy and recording techniques
- `aspect` - Variable accuracy and recording techniques
- `rain_gauge_amount` - We recommend using the BioClim variables to estimate mean annual precipitation
- `catenal_position` - Subject to human bias
- `canopy_cover` - Variable accuracy and recording techniques
- `local_community_rights` - Subject to human bias, sparse data
- `other_user_rights` - Subject to human bias, sparse data
- `timber_harvesting` - Subject to human bias, sparse data
- `high_graded_100_years` - Subject to human bias, sparse data
- `fuel_wood_harvesting` - Subject to human bias, sparse data
- `charcoal_harvesting` - Subject to human bias, sparse data
- `other_woody_product_harvesting` - Subject to human bias, sparse data
- `ntfp_harvesting` - Subject to human bias, sparse data
- `farmed_30_years` - Subject to human bias, sparse data
- `fire_exclusion_success` - Subject to human bias
- `elephant_density` - Variable accuracy

Additionally, SEOSAW takes the stance that metadata should be left blank if it is not known, rather than guessed at. This means that many metadata columns consist mainly of NA values. For example, `charcoal_harvesting` is left mostly blank because most data originators did not record whether wood collection for charcoal harvesting had occurred in the plot. This also applies to stem data. If for example, the data originator did not specify whether a stem was standing or fallen, `standing_fallen` is left as NA, even if it is more likely that the stem was in fact standing.

4 Contributing data

If you would like to contribute data to the SEOSAW network, it is recommended that you format your data in two tables, one for plots and the other for stems, so it is compatible with the existing SEOSAW dataset. The guidance below details the contents of these two tables, and other metadata.

We provide blank template field data sheets in the SEOSAW Bitbucket repository which you can use when setting up a new plot or re-censusing an existing plot (`blank_plot.pdf`, `blank_stem.pdf`). Additionally, we have template data entry files (`plot_data_template.csv`, `stem_data_template.csv`) and a table which will help translate values in the template field data sheets to columns in the template data entry files (`table-match.csv`). Finally, we can provide [ODK forms](#) to allow for data collection on an Android device, with helpful hints on data collection protocol.

4.1 Submission guidelines

Please submit tabular data as `.csv` (comma-separated values) files if possible. If working in Microsoft Excel or a similar spreadsheet program, there is usually an option to export as `.csv`. `.csv` files are preferred as they are simple and store data exactly as entered. `.xls` or `.xlsx` files are also acceptable.

Spatial data such as plot outlines are preferred as shapefiles (`.shp`), though plot corner or plot centre locations as latitude-longitude coordinates, stored in a `.csv` file are also acceptable. Make sure to describe the Coordinate Reference System (CRS) used for coordinates, WGS84 Latitude-Longitude (EPSG: 4326) is preferred.

If you are not using the SEOSAW template data entry files, please provide a description of the contents of each file in a separate document. For tabular data, try to provide a description of the contents of each column. For example, to describe the contents of a column in the stems table containing stem diameter measurements:

S diam - Stem diameters, measured in centimetres to one decimal place (i.e. to the nearest mm).

Provide a description of the methodology used to collect the data in a separate document, especially any plot nested-ness or stratified sampling, as this will determine whether any corrections should be made when calculating plot-level statistics like aboveground biomass.

4.2 Vital data

There are many columns which can be filled by data contributors in the stem and plot tables, and you should try to provide data for as many as possible. We understand however, that some data simply don't exist depending on the original goal of your data collection, Our stance is that metadata should be left blank if it is not known, rather than guessed at.

When compiling your own data, please use the template data entry tables where possible.

Below is a list of the columns which are absolutely necessary for minimum data quality. Data submitted without all of these columns cannot be added to the SEOSAW dataset. The column names as they appear in the template data tables are shown in brackets:

Plots table:

- Plot name, for matching with plot names in stems table (`plot_name`)
- Plot longitude and latitude coordinates (`longitude_of_centre`, `latitude_of_centre`)
- Plot shape and dimensions (`plot_shape`, `plot_length`, `plot_width`, `plot_diameter`)

- The Primary Investigator of the data (`prinv`)

Stems table:

- Plot name, for matching with plot names in plots table (`plot_name`)
- Tree species name (`species_orig_binom` or `species_orig_local`)
- Tree diameter (`diam`)
- Height of diameter measurement (`pom`)
- Fractional Probability of Inclusion. If plots are nested with varying sampling effort within plots for stems of different size, the probability of a stem this size occurring in the whole plot (`fpc`)
- Is the stem alive or dead? (`alive`)

If only a few entries from each of these columns is missing due to mistakes during fieldwork, don't worry, you data is probably still usable, but try to fill as many empty values as possible.

4.3 Additional data

Please provide *any* metadata you think might be even slightly useful to gain a more detailed understanding of your plots. For example, if you have meteorological data for a nearby weather station, herbivore dung counts, photos of the plots, land use history timelines, soil core profiles, or fire management notes. As with the plot and stem data, try to provide a detailed description of the data, the methodology used to collect the data, and any caveats.

5 Cleaning data for the SEOSAW dataset

`{seosawr}` is a package for the SEOSAW dataset. The package serves two main purposes:

1. Formatting data for inclusion in the SEOSAW dataset, or for comparison with the existing SEOSAW dataset.
2. Providing convenience functions for further analysis and manipulation of the SEOSAW dataset.

Most of the functions in the `{seosawr}` R package act on either one of or both the SEOSAW dataset tables, with the column names necessary for that function specified by additional arguments in the function. For example, the function to calculate aboveground biomass:

```
mstems$agb <- agbGen(x = mstems, plot_data = mplots,
  species_name = "species_orig_binom",
  diam = "diam",
  stem_plot_id = "plot_id",
  plot_plot_id = "plot_id",
  latitude_of_centre = "latitude_of_centre",
  longitude_of_centre = "longitude_of_centre",
  allometry = "chave14")
```

The column name arguments default to column names specified by the SEOSAW dataset, but can be altered by the user to allow for some flexibility.

Most top level functions specify a dataframe as the first argument (**x**). Depending on the function, the output is normally either a modified version of **x** with multiple columns added, or a simple vector with the same length as **nrow(x)** and the same row order as **x**. See the **agbGen()** example above for a function which returns a vector, and the species generator function **speciesGen()** for an example which returns a modified version of the stems table:

```
mstems_species <- speciesGen(mstems,
  return_unknown = TRUE, return_unknown_cjb = TRUE, return_unknown_gnr = TRUE)
```

Read the data collection manuals before setting up new plots and collecting data in the field. Follow their protocol to maximise compatibility of newly collected data with existing SEOSAW dataset. The SEOSAW manuals can be found [here](#).

It is suggested that wherever possible you refrain from directly editing raw spreadsheets collected in the field. Instead, use R to create a reproducible script which transforms the raw data into a format which is compatible with the larger SEOSAW dataset. Editing spreadsheets directly is likely to introduce more errors due to mis-typing, and your future self will be unsure of the exact data cleaning procedure, which may be important for future analysis and scientific writing.

The following example outlines a generic data cleaning process for a small fictional dataset of two plots, each with 50 stems each (**mplots_bad**, **mstems_bad**, [Table 3](#), [Table 4](#)). Real datasets are likely to be much larger than this, but the cleaning process should be functionally similar. This is the same cleaning process which is followed for all datasets in the SEOSAW dataset.

Table 3: Example plots table before cleaning.

	plotcode	plot_cluster	plot_name	country_iso3	year	last_census	all_dates	pi	other_authors
1	ABC_1	ABC_1	bicuar_1ha_1	AGO	2019	NA	NA	Godlee, J. L.	Goncalves, F. M., Chisingui, A. V.
2	ABC_2	ABC_1	bicuar_1ha_2	AGO	2019	NA	NA	Godlee, J. L.	Goncalves, F. M., Chisingui, A. V.
3	ABC_3	ABC_1	bicuar_1ha_3	AGO	2019	NA	NA	Godlee, J. L.	Goncalves, F. M., Chisingui, A. V.
4	ABC_4	ABC_1	bicuar_1ha_4	AGO	2019	NA	NA	Godlee, J. L.	Goncalves, F. M., Chisingui, A. V.
NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

Table 4: Example stems table before cleaning.

plotcode	subplot_id	tree	stem	tag_id	measurement_date	measurement_id	species_orig_binom	species_orig_local
ABC_1	NA	1	1	1	2020	NA	Strychnos cocculoides	NA
ABC_1	NA	2	1	2	2020	NA	Androstachys johnsonii	NA
ABC_1	NA	3	1	3	2020	NA	Marquesia macroura	NA
ABC_1	NA	5	1	4	2020	NA	Diplorhynchus condylocarpon	NA
ABC_1	NA	5	2	5	2020	NA	Diplorhynchus condylocarpon	NA

See [Figure 9](#) for a flow diagram of the recommended cleaning process, which outlines the functions used and their order.

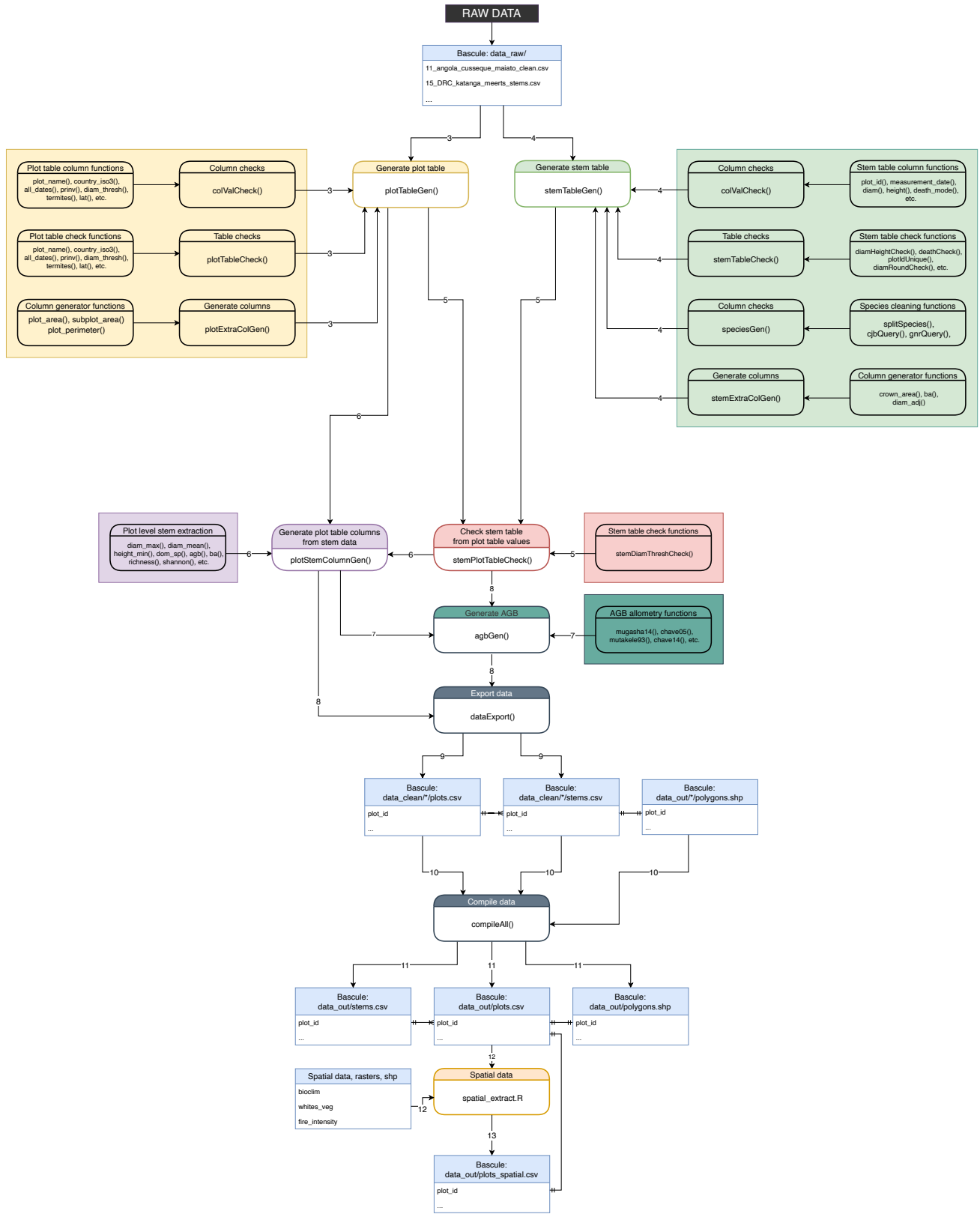


Figure 9: Flow diagram of dataset cleaning.

5.1 Cleaning plot data

5.1.1 Rename columns, replace values

Rename, remove and create columns based on the accepted names found in `plotColClass()`. This function contains a named list, where the names of each list item are the names of columns in the SEOSAW dataset, and the list item is a column constructor function which provides checks and conservatively fixes common errors. While the `{dplyr}` package isn't necessary for this workflow, it excels at dataframe manipulation and makes some parts of the process much more concise to write and easier to interpret by the reader. The `colGen()` function can be used to generate empty columns to remind you which data fields should be provided:

```
plots_clean <- mplots_bad %>%
  dplyr::rename(
    plot_id = plotcode,
    prinv = pi,
    first_census = year) %>%
  dplyr::select(
    -water_table,
    -soil_carbon) %>%
  colGen(., colClass = plotColClass())
```

Then, replace some values which are incompatible with the SEOSAW dataset's column rules:

```
plots_clean_val <- plots_clean %>%
  mutate(
    plot_shape = case_when(
      plot_shape == "square" ~ "rectangle",
      TRUE ~ plot_shape),
    min_diam_thresh = case_when(
      min_diam_thresh == "<5cm" ~ "5",
      min_diam_thresh == "<10cm" ~ "10",
      TRUE ~ min_diam_thresh),
    last_census = first_census,
    all_dates = first_census)
```

Use `colValCheck()` to check whether column names and contents are SEOSAW compatible. This checking function can be used iteratively with the cleaning process above until the function returns no errors. Note that `colValCheck()` returns some warnings and some errors:

```
plots_check <- colValCheck(plots_clean_val, colClass = plotColClass())
```

5.1.2 Check column contents are logically consistent

Next, check that columns in the plots table are consistent with each other using `plotTableCheck()`. This function runs a number of checking functions to ensure that values are logically consistent between columns. For example, `fireCheck()` ensures that the columns `fire_exclusion`, `fire_exclusion_success`, `fire_treatment`, `fire_treatment_season` and `fire_return_interval` are consistent, e.g. if `fire_treatment = TRUE`, `fire_treatment_season` should also contain a non-NA value. As there are always exceptions to perfect data collection practise, these functions return warnings rather than errors, indicating which rows are logically inconsistent. For the default checking functions (`nestedCheck()`, `paCheck()`, `shapeCheck()`, `fireCheck()`, `isoCheck()`) if `output = TRUE` a list of row index vectors are returned, which can then be used to subset the original data to

locate potential errors. The user is encouraged to consider the warnings and check whether these columns are legitimately inconsistent, or if there has been an error during data entry.

```
plotTableCheck(plots_check, output = TRUE)
```

```
## [[1]]  
## list()  
##  
## [[2]]  
## [[2]][[1]]  
## integer(0)  
##  
##  
## [[3]]  
## list()  
##  
## [[4]]  
## list()  
##  
## [[5]]  
## [[5]][[1]]  
## integer(0)
```

The default checking functions, and their checks, are shown below:

- `nestedCheck()`
 - If a nesting level is FALSE (e.g. `nested_1 = FALSE`), make sure nesting descriptors (e.g. `n1_min_diam_thresh`) are NAs, and vice versa.
 - If a lower level of nesting is TRUE (e.g. `nested_2 = TRUE`), make sure higher levels are also TRUE (e.g. `nested_1 = TRUE`).
- `paCheck()`
 - If a plot is in a protected area (i.e. `pa = TRUE`), make sure protected area name (`pa_name`) and type (`pa_type`) are filled in.
- `shapeCheck()`
 - If a plot is a circle or rectangle (e.g. `plot_shape = "circle"`), are the corresponding plot dimension columns filled (i.e. `plot_diameter`, `plot_length`, `plot_width`).
- `fireCheck()`
 - If a plot has a fire treatment or fire exclusion protocol (e.g. `fire_treatment = TRUE` or `fire_exclusion = TRUE`), are the descriptor fields for those treatments filled (e.g. `fire_treatment_season`, `fire_treatment_success`).
- `isoCheck()`
 - Does the country described in `country_iso3` agree with the plot coordinates provided in `longitude_of_centre` and `latitude_of_centre`.

Finally, generate extra columns in the plot level dataset using `plotExtraColGen()`, which calculates values such as `plot_area`, using `plot_shape`, `plot_diameter`, `plot_length` and `plot_width` which have been entered by the user.

```
plots_table <- plotExtraColGen(plots_check)
```

The default functions called by `plotExtraColGen()` are:

- `plot_area()` - Generate the area of the plot in hectares using `plot_shape`, `plot_diameter`, `plot_length` and `plot_width`
- `subplot_area()` - Generate the area of each subplot, in hectares, using `subplot_shape`, `subplot_diameter`, `subplot_length` and `subplot_width`
- `plot_perimeter()` - Generate the perimeter of the plot in metres using `plot_shape`, `plot_diameter`, `plot_length` and `plot_width`

5.1.3 Plot polygons

Generate plot polygons either using plot corner or plot centre coordinates, or estimate the polygon using the plot centre coordinates using the `polyGen()` function and the values in `plot_shape`, `plot_width`, `plot_length`, `plot_diameter`, `latitude_of_centre`, `longitude_of_centre`:

```
plot_polys <- polyGen(plots_table[plots_table$polygons == FALSE,])
```

5.2 Cleaning stem data

5.2.1 Rename columns, replace values

At this point, it is recommended to clean the stem data, using a similar procedure to above:

```
stems_clean <- mstems_bad %>%
  dplyr::rename(plot_id = plotcode,
    tree_id = tree,
    stem_id = stem) %>%
  mutate(height = height / 100,
    alive = case_when(
      alive == "alive" ~ "a",
      alive == "dead" ~ "d",
      TRUE ~ alive),
    fpc = 1,
    diam_method = "dbh_tape")

stems_clean_check <- seosawr::colValCheck(stems_clean,
  colClass = seosawr::stemColClass())
```

For consistency between datasets, it is necessary to simplify and clean `stem_id` and `tree_id` values using the `idClean()` function:

```
stems_clean_check <- idClean(stems_clean_check)
```

See [Table 5](#) for an example of a table with unformatted stem IDs, and [Table 6](#) for an example of the changes `idClean()` makes, showing some of its behaviours. Note that `tag_id` should be reserved for physical tags that allow re-measurement of permanent plots, and should not be altered.

Table 5: Example ID columns before cleaning.

plot_id	tree_id	stem_id	measurement_date
ABC_1	Aa1	Aa1_1	2019
ABC_1	Aa1	Aa1_2	2019
ABC_1	Aa3	NA	2019
ABC_2	Aa2	1	2019
ABC_2	Aa	Aa	2019
ABC_2	NA	1	2019
ABC_2	NA	2	2019
ABC_2	Aa3	3	2019
ABC_2	Aa3	2	2019
ABC_2	Aa3	3	2020

Table 6: Example ID columns after cleaning with `idClean()`.

plot_id	tree_id	stem_id	measurement_date	measurement_id
ABC_1	ABC_1_1	ABC_1_1_1	2019	ABC_1_1_1_2019
ABC_1	ABC_1_1	ABC_1_1_2	2019	ABC_1_1_2_2019
ABC_1	ABC_1_2	ABC_1_2_1	2019	ABC_1_2_1_2019
ABC_2	ABC_2_1	ABC_2_1_1	2019	ABC_2_1_1_2019
ABC_2	ABC_2_2	ABC_2_2_1	2019	ABC_2_2_1_2019
ABC_2	ABC_2_3	ABC_2_3_2	2019	ABC_2_3_2_2019
ABC_2	ABC_2_3	ABC_2_3_1	2019	ABC_2_3_1_2019
ABC_2	ABC_2_3	ABC_2_3_2	2020	ABC_2_3_2_2020
ABC_2	ABC_2_4	ABC_2_4_1	2019	ABC_2_4_1_2019
ABC_2	ABC_2_5	ABC_2_5_1	2019	ABC_2_5_1_2019

5.2.2 Check column contents are logically consistent

Next check for logical inconsistencies between columns in the stem data:

```
stemTableCheck(stems_clean_check)
```

```
## [[1]]
## NULL
##
## [[2]]
## NULL
##
## [[3]]
## NULL
##
## [[4]]
## NULL
```

The default checking functions for `stemTableCheck()`, and their checks, are shown below:

- `diamHeightCheck()`

- Uses a linear model to flag diameter:height ratios that are clearly unusual, based on a σ standard deviation threshold specified by the user.
- `deathCheck()`
 - If a stem is registered as dead (i.e. `alive = "D"`), is the necessary death information provided?
 - If a stem is recorded as damaged by humans (i.e. `damage_cause = "H"`), is `damage_cause_human` filled, and vice versa?
 - If a stem is recorded as broken (i.e. `stem_mode = "P"`), is `broken_height` filled?
- `diamDecCheck()`
 - Identifies diameter values which are unusually common. The function bins diameter measurements and runs a linear regression on the ranked diameter bin frequencies, then checks if any bin frequencies are outside a σ standard deviation threshold.

If the stem data is from the re-measurement of a previously measured plot, you will want to match up the tree IDs and stem IDs with the older dataset, using the tag IDs as a way to match the two datasets. To make this process easier, use `tagTreeStemIDMatch()`:

```
tagTreeStemIDMatch(new_data, old_data,
  new_plot_id = "plot_id", new_tree_id = "tree_id",
  new_stem_id = "stem_id", new_tag_id = "tag_id",
  old_plot_id = "plot_id", old_tree_id = "tree_id",
  old_stem_id = "stem_id", old_tag_id = "tag_id")
```

5.2.3 Fix species names

Next fix the species names. `speciesGen()` generates clean species names by querying both CJB and `taxize::gnr_resolve()`. Firstly the function checks whether a species has been observed in the larger SEOSAW dataset. Given that the SEOSAW dataset contains in excess of 500,000 stem measurements, if a species is new to the dataset it is likely to be a mistake. Secondly the function checks `taxize::gnr_resolve()` using the `gnrQuery()` wrapper function, and thirdly it checks the CJB database using `cjbQuery()`. At any of these three steps it is possible for the user to specify that the function should return unmatched species names. This is especially useful when cleaning a new dataset where species names may be misspelled or misclassified and need to be manually corrected.

There are a number of lookup tables provided as data objects in the `{seosawr}` package which deal with synonymy issues that aren't handled appropriately by CJB and/or `taxize::gnr_resolve()`:

- `data(acacia)`
- `data(maytenus)`
- `data(trillesanthus)`
- `data(synonymy)`
- `data(rhus)`

```
stems_species <- speciesGen(stems_clean_check,
  return_unknown = TRUE, return_unknown_cjb = TRUE, return_unknown_gnr = TRUE)
```

If species names are not found, `stems_species` will contain a vector of species names not matched. The user can then use this vector to create a lookup table which corrects the species names. If you are sure the species name is correct, just replace the old name with the same name in the lookup table:

```
lookup_table <- data.frame(
  original = stems_species,
  corrected = c("Lankea discolor", "Marquesia macroura", "Syzygium cordatum"))
```

Then run `speciesGen()` for a second time, this time passing the lookup table as an argument. This will replace the species name with the corrected version before they are passed to the checking routines.

```
stems_species <- speciesGen(stems_clean_check,
  return_unknown = FALSE, return_unknown_cjb = FALSE, return_unknown_gnr = FALSE,
  lookup_ortho = lookup_table)
```

If `speciesGen()` is successful, `stems_species` will now be filled with the stem level dataframe with extra columns appended by `speciesGen()`.

The function `abMatGen()` can be used to create an abundance matrix from a stem measurements table. An abundance matrix is a table where each column is a species, each row is a plot, and each cell contains abundance values for each species in each plot. The most basic use of `abMatGen()` uses tree abundance as a measure of abundance, but other columns can be used as alternative abundance measures, such as aboveground biomass (agb) or basal area (ba). The function also accounts for variation in sampling effort across nested plots using the `fpc` columns.

```
abMatGen(mstems, plot_id = "plot_id", tree_id = "tree_id",
  species_name = "species_orig_binom", fpc = "fpc")[,1:4]
```

```
##      Androstachys johnsonii Balanites maughamii
## ABC_1                3                1
## ABC_2                3                3
## ABC_3                3                4
## ABC_4                5                2
##      Baphia massaiensis Bauhinia petersiana
## ABC_1                0                4
## ABC_2                1                0
## ABC_3                1                1
## ABC_4                1                1
```

5.2.4 Generate new columns

Next, add extra stem columns using `stemExtraColGen()`:

```
stems_extra <- stemExtraColGen(stems_species)
```

`stemExtraColGen()` runs a number of functions to create new columns:

- `crown_area()` - Crown area modelled as an ellipse if both `crown_x` and `crown_y` are provided, otherwise modelled as a circle if only one provided, in m².
- `ba()` - Stem basal area in square metres, calculated using DBH measurements (`diam`), assuming a circular trunk cross-section.

- `diam_adj()` - If the DBH was taken at a POM (pom) other than 1.3 m, an estimated DBH at 1.3 m, assuming a degree of stem taper with height.

Calculate aboveground biomass using the Chave, Réjou-Méchain et al. (2014) `chave14()` allometric equation, which requires both plot level longitude (`longitude_of_centre`) and latitude (`latitude_of_centre`) as well as stem diameter (`diam`) and species names. Use `species_name_clean` which was generated by `speciesGen()` rather than the original `species_orig_binom` if calculating AGB at this stage, to ensure that wood density estimates are correct.

```
stems_extra$agb <- agbGen(stems_extra, plots_table,
  species_name = "species_name_clean")
```

As a side note, the wood density data used by `chave14()` can be generated separately using `woodDensityQuery()`:

```
head(woodDensityQuery(stems_extra, species_name = "species_orig_binom",
  plot_id = "plot_id", region = "AfricaTrop"))
```

```
## [1] 0.7175 0.9040 0.9040 0.7090 0.7090 0.7800
```

There are a number of other allometric equations available in the `{seosawr}` package which the user can use at their discretion (Table 7). There is also the option for the user to use their own allometric equation on the SEOSAW dataset.

The SEOSAW dataset contains `local_allometry` column in the plots table which allows data originators to suggest their own local allometry, though these allometries aren't available as functions in the `{seosawr}` package as they are numerous.

Function	Inputs	Usage	Reference
<code>ryan11()</code>	Diameter	Generic miombo woodland	(Ryan, Williams and Grace, 2011)
<code>mugasha14()</code>	Diameter	Tanzanian miombo woodland	(Mugasha et al., 2013)
<code>chidumayo90()</code>	Diameter	Zambian woodlands	(Chidumayo, 1990)
<code>mutakela09()</code>	Diameter	Mopane woodlands in Botswana	(Mutakela, 2009)
<code>chamshama04()</code>	Diameter	Tanzanian miombo woodlands	(Chamshama, Mugasha and Zahabu, 2004)
<code>tietema93()</code>	Diameter	Mopane woodlands in Botswana	(Tietema, 1993)
<code>chave05()</code>	Wood density Height Diameter	Global dry forests	(Chave, Andalo et al., 2005)
<code>chave14()</code>	Wood density Latitude Longitude Diameter	Global dry forests	(Chave, Réjou-Méchain et al., 2014)

Table 7: The various allometric equations to estimate aboveground woody biomass (AGB) provided in the `{seosawr}` R package.

The `{seosawr}` package relies on functions from the `{BIOMASS}` package to calculate the `chave14()` and `chave05()` allometries, which uses wood density data from (Chave, Coomes et al., 2009). Aboveground woody biomass can be calculated using the `agbGen()` function, which returns aboveground biomass per stem in tonnes of dry matter.

Wood density data used by `chave14()` and `chave05()` can be generated separately using `woodDensityQuery()`, which is a wrapper around `BIOMASS::getWoodDensity()`.

5.2.5 Check logical consistency of plots and stems tables

Next, we are in a position to compare the logical consistency of the stem table with the plot table. Much like `stemTableCheck()` and `plotTableCheck()`, this function conducts a number of checking procedures and flags rows in the stems table which don't agree with the plot level data, such as if a stem diameter is smaller than the minimum diameter threshold recorded for the plot.

```
stemPlotTableCheck(stems_extra, plots_table)
```

The functions run by `stemPlotTableCheck()` are:

- `stemDiamThreshCheck()` - Are there stem measurements outside of the minimum or maximum stem diameter thresholds. Note that if a plot is nested with diameter thresholds, the highest of these thresholds is used and so may return warnings for stems in nested subplots.
- `plotIDMatchCheck()` - Check if all plot IDs are present in both tables.

5.2.6 Generate plot level statistics from stem data

Then, you can calculate plot level statistics from the stems table and add them to the plots table. This includes calculations like the plot level aboveground biomass, the number of stems at different size thresholds, and the most dominant species in each plot:

```
plots_table_stat <- plotStemColumnGen(stems_extra, plots_table)
```

The statistics calculated are:

- `diam_max` - Maximum stem diameter
- `diam_min` - Minimum stem diameter
- `diam_mean` - Mean stem diameter
- `diam_median` - Median stem diameter
- `diam_sd` - Standard deviation of stem diameter distribution
- `height_max` - Maximum height
- `height_min` - Minimum height
- `height_mean` - Mean height
- `height_median` - Median height
- `height_sd` - Standard deviation of height
- `n_stems` - Number of stems
- `n_diam` - Number of stems with diameter measurements
- `n_height` - Number of stems with height measurements
- `lorey_height` - Lorey height
- `ba` - Total basal area

- `agb` - Total aboveground biomass
- `n_determ_genus` - Number of successfully determined genera
- `n_determ_species` - Number of successfully determined species

Many of the above are replicated with subsets of data at various minimum stem diameter thresholds: `>0`, `>5`, `>10`, `>20`, `>40`, `>60` cm.

Additionally, diversity statistics are calculated using `diversityGen()` and the dominant species are identified using `domSpeciesGen()`:

- `richness` - Species richness
- `rarefy` - Rarefied species richness
- `shannon` - Shannon-Wiener diversity index H'
- `simpson` - Simpson diversity index D
- `invsimpson` - Inverse Simpson diversity index
- `evenness` - Species abundance evenness

A number of statistics are recalculated as per-hectare variants:

- `n_stems_ha`
- `ba_ha`
- `agb_ha`

If you want to add any column from the plot data into the stem data, use `plotToStem()`:

```
mstems_analysis <- plotToStem(stems_extra, plots_table_stat, cols = c("agb_ha", "ba_ha"),
  stem_plot_id = "plot_id", plot_plot_id = "plot_id")
```

FINALLY, the data is now in the SEOSAW format, with all metadata attached and some preliminary analysis already completed. You can export data using the `exportData()` function, or using the normal `write.csv()`, `st_write()` etc.

```
exportData("data_clean/example_data", stems = stems_extra,
  plots = plots_table_stat, polys = plot_polys, overwrite = FALSE,
  files = c("stems_csv", "plots_csv", "points_shp"))
```

5.3 Compiling datasets

Once multiple datasets have been cleaned according to the process outlined above, they can be compiled to a single dataset using `compileAll()`:

```
compileAll(dirs = c("data_1", "data_2"),
  files = c("stems_csv", "plots_csv", "points_shp",
    "polys_shp"), out_dir = "../path/to/out/dir")
```

There are also two functions for extracting subsets of a compiled dataset. The first, `subsetLatest()` extracts the most recent census data for each plot, and `subsetMulti()` extracts measurements only for those plots which have multiple censuses:

```
subsetLatest(compiled_data, plot_id = "plot_id", diam = "diam",
  measurement_date = "measurement_date", alive = "alive")
```

`versionCheck()` allows you to check for differences in versions of a dataset. This is mostly used internally when creating the SEOSAW dataset, but might be useful elsewhere. The function takes two tables of plot data, and a vector of column names within which to check the values of each plot:

```
version_diff <- versionCheck(new_plots, old_plots, cols = c("agb", "n_stems"))
```

The `{seosawr}` package contains generic functions for extracting plot level data from arbitrary raster (`rasterExtract()`) and vector (`vectorExtract()`) spatial objects.

`rasterExtract()` can be used to extract field values from other user supplied raster datasets, with an optional circular buffer around the point of interest. `rasterExtract()` also allows for extraction from factor rasters using the `Mode()` function, which returns the modal value from raster cells within the buffer radius.

`vectorExtract()` is a similar function for extracting attributes from spatial vector datasets for points of interest.

```
africa <- get(data(africa))

mplots_africa <- vectorExtract(mplots, africa, attrib = "iso3")

head(mplots_africa)

##   plot_id iso3
## 1  ABC_1 AGO
## 2  ABC_2 AGO
## 3  ABC_3 AGO
## 4  ABC_4 AGO
```

The `{seosawr}` package also contains various convenience functions for manipulating spatial data. `{seosawr}` primarily uses `{sf}` spatial data objects, as this is likely the future of spatial analysis in R, but some functions such as `reprojectPoints()` will alternatively return a `SpatialPoints` object if requested.

A common scenario is that data originators like to store their plot coordinates as cartesian UTM coordinates for analysis in their local region, but SEOSAW requires all coordinates as WGS84 decimal latitude/longitude values (EPSG: 4326). As an example, imagine a dataset where plot coordinates straddle both UTM 37S and UTM 38S. We want to convert the plot coordinates to WGS84 latitude/longitude:

```
head(mplots[,c("plot_id", "longitude_of_centre", "latitude_of_centre")])

##   plot_id longitude_of_centre latitude_of_centre
## 1  ABC_1             14.79         -15.17
## 2  ABC_2             14.79         -15.16
## 3  ABC_3             14.79         -15.17
## 4  ABC_4             14.78         -15.17
```

```

mplots$utm_zone <- latLong2UTM(mplots$longitude_of_centre, mplots$latitude_of_centre)

mplots_list <- split(mplots, mplots$utm_zone)

utm_crs <- unlist(lapply(mplots_list, function(x) {
  UTMProj4(unique(x$utm_zone))
})))

mplots_list_wgs <- lapply(1:length(mplots_list), function(x) {
  reprojectPoints(mplots_list[[x]]$longitude_of_centre,
    mplots_list[[x]]$latitude_of_centre,
    from = utm_crs[[x]], to = 4326)
})

mplots_wgs <- do.call(rbind, mplots_list_wgs)

head(mplots_wgs[,c("plot_id", "longitude_of_centre", "latitude_of_centre")])

```

6 Dataset table columns

This section provides a comprehensive description of every column found in the SEOSAW dataset, across the stems table, plots table, and spatial data table.

Table 8: Column names for plots table.

Column	Class	Description	Units	Example
plot_id	chr	A standardised method of identifying plots. For example, the first plot in a set of plots installed in Angola (A), in Bicuar National Park (B), by John Godlee (G), would be referred to as ABG_1. The three letter code should be conserved within a dataset, i.e. a group of plots owned by a particular researcher within a particular locale.	NA	SSM_1
plot_cluster	chr	If plots occur in a spatial cluster, a single plot_id used to identify the cluster	NA	ABG_1
plot_name	chr	The plot name as designated by those who originally collected the data. 1, Wedza_01, MHR-04b are all valid. No spaces please.	NA	Wedza_01
country_iso3	fct	The ISO 3166-1 alpha-3 code to refer to the country in which the plot is situated. Only countries within Africa. See data(africa_iso3) for available country codes. use countrycode::countrycode() to convert a country name to an iso_3 code	NA	AGO
first_census	chr	The year of the first plot census	NA	1993
last_census	chr	The year of the most recent plot census	NA	2019
all_dates	chr	A list of all the years censuses were taken, separated by commas in a single character string	NA	2011, 2012, 2019
prinv	chr	The name of the Principal Investigator. This can be an organisation, a person's name, a surname, initials, etc.	NA	Godlee, J. L.
other_authors	chr	The names of other authors who contributed to the collection of the dataset, separated by commas	NA	Godlee, J. L., Maiato, F. P., Dexter, K.
plot_shape	fct	The shape of the plot. Possible values are: "rectangle", "circle" or "irregular". Note that a square is a special case of "rectangle".	NA	rectangle
plot_width	int	If plot_shape == "rectangle", the shorter side length, in metres to the nearest metre	m	100
plot_length	int	If plot_shape == "rectangle", the longer side length, in metres to the nearest metre	m	50
plot_diameter	int	If plot_shape == "circle", the diameter of the plot, measured in metres to the nearest metre	m	25
subplot_n	int	Number of subplots within the larger plot. Subplots (as opposed to nested plots) have the same sampling effort as the entire plot, generally used for ease of relocating trees.	n	10
subplot_shape	fct	The shape of the subplots. Possible values are the same as plot_shape	NA	circle
subplot_length	int	If subplot_shape == "rectangle", the longest side length, in metres to the nearest metre	m	50

Table 8: Column names for plots table. *(continued)*

Column	Class	Description	Units	Example
subplot_width	int	If subplot_shape == "rectangle", the shorter side length, in metres to the nearest metre	m	10
subplot_diameter	int	If subplot_shape == "circle", the diameter of the plot, measured in metres to the nearest metre	m	10
min_diam_thresh	int	The minimum stem diameter value considered within this plot, in centimetres to the nearest centimetre. If sampling is nested, the minimum value which is surveyed across the entire plot area.	cm	5
max_diam_thresh	int	The maximum stem diameter value considered within this plot, in centimetres to the nearest centimetre. Note if no maximum stem diameter then NA. If sampling is nested, the maximum value which is considered across the entire plot area.	cm	100
min_height_thresh	num	The minimum stem height value considered within this plot, in metres to the nearest 10 centimetres. If sampling is nested, the minimum value which is surveyed across the entire plot area.	m	1.5
max_height_thresh	num	The maximum stem height value considered within this plot, in metres to the nearest 10 centimetres. If sampling is nested, the maximum value which is surveyed across the entire plot area.	m	50.5
nested_1	logi	Does the plot have at least one level of nested sampling?	NA	TRUE
n1_min_diam_thresh	int	If nested_1 == TRUE, the minimum stem diameter value considered within the first nesting level, in centimetres to the nearest centimetre.	cm	10
n1_min_height_thresh	num	If nested_1 == TRUE, the minimum stem height value considered within the first nesting level, in metres to the nearest metre.	m	20
n1_plot_shape	fct	If nested_1 == TRUE, the shape of the subplot, same values as plot_shape	NA	circle
n1_plot_width	int	If nested_1 == TRUE & n1_plot_shape == "rectangle" the shorter side length, in metres to the nearest metre	m	20
n1_plot_length	int	If nested_1 == TRUE & n1_plot_shape == "rectangle" the longer side length, in metres to the nearest metre	m	10
n1_plot_diameter	int	If nested_1 == TRUE & n1_plot_shape == "circle" the diameter of the subplot, in metres to the nearest metre	m	14
nested_2	logi	Does the plot have at least two levels of nested sampling?	NA	TRUE
n2_min_diam_thresh	int	If nested_2 == TRUE, the minimum stem diameter value considered within the second nesting level, in centimetres to the nearest centimetre.	cm	5

Table 8: Column names for plots table. (*continued*)

Column	Class	Description	Units	Example
n2_min_height_thresh	num	If nested_2 == TRUE, the minimum stem height value considered within the second nesting level, in metres to the nearest metre.	m	2.5
n2_plot_shape	fct	If nested_2 == TRUE, the shape of the subplot, same allowed values as plot_shape	NA	rectangle
n2_plot_width	int	If nested_2 == TRUE & n2_plot_shape == "rectangle" the shorter side length, in metres to the nearest metre	m	30
n2_plot_length	int	If nested_2 == TRUE & n2_plot_shape == "rectangle" the longer side length, in metres to the nearest metre	m	20
n2_plot_diameter	int	If nested_2 == TRUE & n2_plot_shape == "circle" the diameter of the subplot, in metres to the nearest metre	m	5
nested_3	logi	Does the plot have at least three levels of nested sampling?	NA	TRUE
n3_min_diam_thresh	int	If nested_3 == TRUE, the minimum stem diameter value considered within the third nesting level, in centimetres to the nearest centimetre.	cm	5
n3_min_height_thresh	num	If nested_3 == TRUE, the minimum stem height value considered within the third nesting level, in metres to the nearest metre.	m	2.5
n3_plot_shape	fct	If nested_3 == TRUE, the shape of the subplot, same allowed values as plot_shape	NA	rectangle
n3_plot_width	int	If nested_3 == TRUE & n3_plot_shape == "rectangle" the shorter side length, in metres to the nearest metre	m	30
n3_plot_length	int	If nested_3 == TRUE & n3_plot_shape == "rectangle" the longer side length, in metres to the nearest metre	m	20
n3_plot_diameter	int	If nested_3 == TRUE & n3_plot_shape == "circle" the diameter of the subplot, in metres to the nearest metre	m	5
lianas_sampled	logi	Were lianas sampled?	NA	TRUE
soil_sampled	logi	Were soil samples collected?	NA	FALSE
understory_sampled	logi	Was the understorey sampled for grasses, forbs, shrubs, or small trees that did not fit the plot criteria above?	NA	TRUE
dead_stems_sampled	logi	Were dead stems sampled in this plot?	NA	FALSE
stumps_sampled	logi	Were tree stumps sampled?	NA	TRUE
tags	logi	Are stems within the plot tagged?	NA	FALSE
permanent	logi	Is the plot a permanent plot (TRUE) or a one-off census (FALSE)?	NA	TRUE
sampling_notes	chr	Could the plot be re-measured in the future? Extra notes relating to the sampling process	NA	Only stumps greater than 2 m tall were sampled

Table 8: Column names for plots table. (*continued*)

Column	Class	Description	Units	Example
sampling_design_code	chr	The file name for a document containing the description of your methods/protocol for data collection, provided to SEOSAW. If not documented then NA	NA	SSAW_V1.gdoc
open_access	logi	Are you happy for the data to be made public on the SEOSAW data platform? If FALSE the data will be limited to you and SEOSAW members, with permission sought prior to use.	NA	FALSE
metadata_public	logi	Are you happy for the plot metadata (plot location, owner, date of collection, etc.) to be made available to the public? If FALSE, talk to Casey to discuss (casey.ryan@ed.ac.uk)	NA	FALSE
plot_notes	chr	Include any long form comments about the plot here, if they don't fit into other columns.	NA	It looks like this plot is a rubbish dump for household waste.
longitude_of_centre	num	The longitude of the centre of gravity of the plot. If you have GPS points of the plot corners, send those and we will calculate this. Measured in decimal degrees, WGS84 EPSG:4326	deg	-50.425622
latitude_of_centre	num	The latitude of the centre of gravity of the plot. If you have GPS points of the plot corners, send those and we will calculate this. Measured in decimal degrees, WGS84 EPSG:4326	deg	30.446453
elevation	num	The elevation of the plot. Measured in metres above International Mean sea level to 1 decimal place (nearest 10 cm). In order of preference: 1) Average of GPS readings across the plot, 2) a single GPS reading, 3) Manual lookup on topographic map, 4) NA. We can query SRTM for missing values.	m asl	1234
slope_method	ftc	The method used to determine plot slope. Allowed values: "rangefinder" if a laser or sonic rangefinder was used, "trig" if slope was estimated with tape measures and-or a clinometer, "eye" if estimated by eye, "other" if another method.	NA	trig
slope_method_notes	chr	Extra notes relating to the slope method	NA	20 points taken with 20 m minimum distance, randomly located
slope	num	The plot slope, measured in degrees from 0 (flat) to 90 (vertical)	deg	4.4
aspect	num	The aspect of the plot, i.e. which direction is the slope facing, in degrees (0-360)	deg	54
polygons	logi	Are polygon data of the plot shape provided by the data originator?	NA	TRUE

Table 8: Column names for plots table. *(continued)*

Column	Class	Description	Units	Example
local_allometry	fct	The most appropriate aboveground woody biomass allometry for the plot. In order of preference: (1) DOI, (2) reference, (3) other URL, shorthand allometry name	NA	ryan11
allometry_notes	chr	Extra notes relating to the appropriate local allometry	NA	ryan11 should be used for everything except the eucalyptus trees present in plot ABC_5
rain_gauge_near	logi	Is there a rainfall gauge within 20 km of the plot which provides multiple decades of rainfall data? TRUE or FALSE	NA	TRUE
rain_gauge_longitude	num	If rainfall_gauge_near == TRUE, the longitude of the location of the rainfall gauge, measured in decimal degrees	deg	-50.425622
rain_gauge_latitude	num	If rainfall_gauge_near == TRUE, the latitude of the location of the rainfall gauge, measured in decimal degrees	deg	30.446453
rain_gauge_amount	num	If rainfall_gauge_near == TRUE, the average yearly rainfall in mm over the available data period. Measured in mm to the nearest mm	mm yr ⁻¹	553
catenal_position	fct	Where on the catena is plot? Possible values are:"ridge_crest", "concave_slope", "midslope", "convex_slope", "valley_floor", "flat"	NA	midslope
termites	int	If termites were counted, the number of termite mounds found on the plot	n	34
canopy_cover	num	If canopy cover was estimated, the proportion of ground covered by a tree canopy. Measured as a proportion between 0 and 1, e.g. 60% = 0.6	prop	25
canopy_cover_method	fct	If canopy cover was estimated, what method was used? Possible values are: "periscope_densitometer", "mirror_densitometer", "hemispherical_photo", "aerial_photo", "canopy_dimensions", "other"	NA	hemispherical_photo
canopy_cover_method_notes	chr	Extra notes relating to the canopy cover method	NA	40 hemispherical photos in a regular grid with 20 m spacing. Analysed with Hemiphot.

Table 8: Column names for plots table. (*continued*)

Column	Class	Description	Units	Example
xy_method	fct	The method which was used to estimate stem location within the plot, either grid or GPS coordinates. Possible values: "tape", "gps", "dgps", "angle" (angular direction and distance from plot centre in circular plot), "other".	NA	tape
manipulation_experiment	logi	Has any aspect of the plot's land use or environment been manipulated by researchers?	NA	FALSE
experiment_ref	chr	If manipulation_experiment == TRUE, a link, citation, or contact details relating to the experimental manipulation of this plot. In order of preference: (1) DOI, (2) reference, (3) other URL, (4) contact details	NA	10.1890/06-1664
manipulation_experiment_notes	chr	If manipulation_experiment == TRUE, notes relating to the experimental treatment applied to the plot	NA	Tilling of soil 15-20 cm throughout
pa	logi	Is the plot in a protected area?	NA	TRUE
pa_name	chr	If pa == TRUE, the name of the protected area. If the protected area is listed by the WDPa, please use the name provided by the WDPa	NA	Bicuar National Parl
pa_type	chr	If pa == TRUE, the type of protected area. If the protected area is listed by the WDPa, use their type classification.	NA	Forest reserve
wdpa_id	chr	If pa == TRUE and the protected area is listed by the WDPa, the WDPa ID value for that protected area	NA	370
local_community_rights	chr	Explain the rights of the local community in using the plot for resource extraction, grazing, farming etc.	NA	Dead wood collection by local village
other_user_rights	chr	Explain the rights of any others in using the plot for resource extraction, grazing, farming etc.	NA	Yearly game drive in early May
timber_harvesting	logi	Is there evidence of tree harvesting for timber products, e.g. planks?	NA	TRUE
high_graded_100_years	logi	Is it likely that high quality timber trees were removed from the plot the last 100 years?	NA	FALSE
fuel_wood_harvesting	logi	Is there evidence of fuel wood collection, excluding charcoal production?	NA	TRUE
charcoal_harvesting	logi	Is there evidence of trees being harvested for charcoal production?	NA	FALSE
other_woody_product_harvesting	logi	Is there evidence of people harvesting woody material for any other use, e.g. string, beehives, canoes?	NA	TRUE
ntfp_harvesting	logi	Is there evidence of people harvesting non-wood tree products, e.g. mushrooms honey, fruit etc.?	NA	FALSE
farmed_30_years	logi	Is it likely that the plot was used for farming (or cleared for another reason), in the last 30 years?	NA	TRUE

Table 8: Column names for plots table. *(continued)*

Column	Class	Description	Units	Example
land_use_notes	chr	Extra notes relating to land use in the plot	NA	FALSE
tree_stem	fct	Have measurement been taken at the tree level or stem level? Possible values: "tree", "stem"	NA	tree
tree_diff	logi	Were measurements differentiated into separate trees?	NA	TRUE
fire_exclusion	logi	Is there a fire exclusion experiment in this plot? If TRUE, manipulation_experiment should also = TRUE.	NA	TRUE
fire_exclusion_success	logi	If fire_exclusion == TRUE, to what extent were fires successfully excluded from the plot? Possible values: "all", "most", "few"	NA	all
fire_treatment	logi	Is there a fire treatment experiment in this plot, e.g. controlled fires every two years?	NA	FALSE
fire_return_interval	int	The number of years between fires. If there has been some sort of fire control experiment, the treatment fire return interval measured in years, e.g. 1 for annual burning. If no treatment, the estimated mean interval between fires.	year	5
fire_treatment_season	fct	If fire_treatment == TRUE, the season in which fires are set. Possible values: "early_dry_season", "late_dry_season", "early_wet_season", "late_wet_season"	NA	early_dry_season
fire_treatment_notes	chr	Extra notes relating to fire treatment regime	NA	Fires are set with drip torches in the late evening, after a heavy rain
elephant_exclosure	logi	Are elephants excluded from this plot?	NA	FALSE
elephants	logi	Do elephants frequent the plot	NA	TRUE
elephant_density	num	If elephants are present, at what approximate density do they occur within the area? Measured as elephants per km ²	n km ⁻²	5
elephant_data_source	chr	A link, citation, or contact details relating to elephant density estimates. In order of preference: (1) DOI, (2) reference, (3) other URL, (4) contact details	NA	10.6088/ijes.002
exclosure	fct	Are browsers excluded from the plot by a fence? If so, which animals? Possible values are: NA, "large" if only large animals excluded, "small" if only small animals excluded, "large_small" if both excluded	NA	large
small_nonsocial_browsers	logi	Do small non-social browsers frequent this plot? e.g. dik-diks, duikers and grysbok, steenbok (these categories are based on Hempson et al 2015). TRUE or FALSE	NA	TRUE
large_browsers	logi	Do large browsers frequent this plot? e.g. giraffe, okapi, kudu	NA	FALSE

Table 8: Column names for plots table. (*continued*)

Column	Class	Description	Units	Example
medium_sized_social_mixed_diet	logi	Do medium sized social mixed diet herbivores frequent this plot? e.g. gazelle, oryx	NA	TRUE
nonruminants_excluding_suids	logi	Do non-ruminant herbivores frequent this plot? e.g. zebra, hippo, rhino	NA	FALSE
water_dependent_grazers	logi	Do water-dependent grazers frequent this plot? e.g. roan, sable, bluebuck, waterbuck, lechwe		TRUE
goat_grazing	logi	Do goats feed in this plot?	NA	TRUE
cattle_grazing	logi	Do domestic cattle feed in the plot?	NA	TRUE
herbivory_notes	chr	Extra notes relating to herbivory	NA	Dik Diks, palankas, Olongo, elephants like to knock over the B. massaiensis
data_ref	chr	A link, citation, or contact details relating to the publication of this plot's data. In order of preference: (1) DOI, (2) reference, (3) other URL, (4) contact details	NA	10.3390/d12040140
file_plots	chr	The filepath of the raw plot metadata file provided to SEOSAW. Must end in .xls(x) or .csv	NA	archibald_kruger.xlsx
file_stems	chr	The filepath of the raw stem data file provided to SEOSAW. Must end in .xls(x) or .csv	NA	Archibald_kruger.csv
plot_area	num	The flat area of the plot, in hectares	ha	0.25
subplot_area	num	The flat area of the subplots, in hectares	ha	0.005
plot_perimeter	num	The perimeter of the plot, in metres	m	400
diam_max	num	The maximum stem diameter found in the plot	cm	50.4
diam_min	num	The minimum stem diameter found in the plot	cm	5
diam_mean	num	The mean stem diameter found in the plot	cm	5.6
diam_median	num	The median stem diameter found in the plot	cm	10.3
diam_sd	num	The standard deviation stem diameter found in the plot	cm	1.5
diam_gt5_max	num	The maximum stem diameter found in the plot, greater than 5 cm diameter	cm	50.4
diam_gt5_min	num	The minimum stem diameter found in the plot, greater than 5 cm diameter	cm	5.1
diam_gt5_mean	num	The mean stem diameter found in the plot, greater than 5 cm diameter	cm	5.4
diam_gt5_median	num	The median stem diameter found in the plot, greater than 5 cm diameter	cm	10.6

Table 8: Column names for plots table. (*continued*)

Column	Class	Description	Units	Example
diam_gt5_sd	num	The standard deviation stem diameter found in the plot, greater than 5 cm diameter	cm	1.5
diam_gt10_max	num	The maximum stem diameter found in the plot, greater than 10 cm diameter	cm	50.4
diam_gt10_min	num	The minimum stem diameter found in the plot, greater than 10 cm diameter	cm	10.2
diam_gt10_mean	num	The mean stem diameter found in the plot, greater than 10 cm diameter	cm	15.7
diam_gt10_median	num	The median stem diameter found in the plot, greater than 10 cm diameter	cm	15.9
diam_gt10_sd	num	The standard deviation stem diameter found in the plot, greater than 10 cm diameter	cm	2.5
n_diam	num	The number of stems for which diameter was measured	n	524
n_height	num	The number of stems for which height was measured	n	532
height_max	num	The maximum stem height found in the plot	m	10.5
height_min	num	The minimum stem height found in the plot	m	1.6
height_mean	num	The mean stem height found in the plot	m	1.6
height_median	num	The median stem height found in the plot	m	5.4
height_sd	num	The standard deviation stem height found in the plot	m	4.5
height_gt5_max	num	The maximum stem height found in the plot, greater than 5 cm DBH	m	10.5
height_gt5_min	num	The minimum stem height found in the plot, greater than 5 cm DBH	m	1.6
height_gt5_mean	num	The mean stem height found in the plot, greater than 5 cm DBH	m	3.5
height_gt5_median	num	The median stem height found in the plot, greater than 5 cm DBH	m	5.4
height_gt5_sd	num	The standard deviation stem height found in the plot, greater than 5 cm DBH	m	1.5
height_gt10_max	num	The maximum stem height found in the plot, greater than 10 cm DBH	m	10.5
height_gt10_min	num	The minimum stem height found in the plot, greater than 10 cm DBH	m	4.5
height_gt10_mean	num	The mean stem height found in the plot, greater than 10 cm DBH	m	3.5
height_gt10_median	num	The median stem height found in the plot, greater than 10 cm DBH	m	6.6
height_gt10_sd	num	The standard deviation stem height found in the plot, greater than 10 cm DBH	m	1.4
n_stems	num	The number of stems in the plot	n	5025
n_stems_gt5	num	The number of stems greater than 5 cm DBH	n	5010

Table 8: Column names for plots table. *(continued)*

Column	Class	Description	Units	Example
n_stems_gt10	num	The number of stems greater than 10 cm DBH	n	2040
n_stems_gt20	num	The number of stems greater than 20 cm DBH	n	505
n_stems_gt40	num	The number of stems greater than 40 cm DBH	n	20
n_stems_gt60	num	The number of stems greater than 60 cm DBH	n	1
lorey_height	num	The lorey height of the plot	NA	3.5
ba	num	Plot level basal area, in m ²	m ²	4.6
agb	num	Plot level aboveground woody biomass, in tonnes of dry matter	t DM	30.5
n_determ_genus	num	Number of stems where genera successfully identified	n	20
n_determ_species	num	Number of stems where species epithets successfully identified	n	49
dom_1	chr	The most dominant species in the plot, by stem abundance	NA	Brachystegia spiciformis
dom_2	chr	The second most dominant species in the plot, by stem abundance	NA	Ochna pulchra
dom_3	chr	The third most dominant species in the plot, by stem abundance	NA	Burkea africana
dom_4	chr	The fourth most dominant species in the plot, by stem abundance	NA	Indet indet
dom_5	chr	The fifth most dominant species in the plot, by stem abundance	NA	Pseudolachnostylis maprouneifolia
prop_1	num	Proportional abundance of the most dominant species in the plot	prop	20.4
prop_2	num	Proportional abundance of the second most dominant species in the plot	prop	14.5
prop_3	num	Proportional abundance of the third most dominant species in the plot	prop	12.7
prop_4	num	Proportional abundance of the fourth most dominant species in the plot	prop	8.4
prop_5	num	Proportional abundance of the fifth most dominant species in the plot	prop	3.6
richness	num	Species richness of plot	n	53
rarefy	num	Rarefied species richness at 25 individuals	NA	42.65
shannon	num	Shannon diversity index of plot	NA	4.3
simpson	num	Simpson diversity index of plot	NA	2.5
invsimpson	num	Inverse Simpson diversity index of plot	NA	2.6
evenness	num	Abundance evenness of plot	NA	2.6
stocking_ha	num	Number of stems per hectare	n ha ⁻¹	50.2
stocking_gt5_ha	num	Number of stems greater than 5 cm DBH per hectare, adjusted for nested sampling with fpc	n ha ⁻¹	45.5
stocking_gt10_ha	num	Number of stems greater than 10 cm DBH per hectare, adjusted for nested sampling with fpc	n ha ⁻¹	32.5

Table 8: Column names for plots table. *(continued)*

Column	Class	Description	Units	Example
ba_ha	num	Basal area per hectare	m ² ha ⁻¹	3.6
agb_ha	num	Aboveground woody biomass per hectare	t DM ha ⁻¹	42.6

Table 9: Column names for stems table.

Column	Class	Description	Units	Example
plot_id	chr	A standardised method of identifying plots. For example, the first plot in a set of plots installed in Angola (A), in Bicuar National Park (B), by John Godlee (G), would be referred to as ABG_1.	NA	ABG_1
subplot_id	chr	If the plot is divided into multiple parcels, but not "nested", i.e. sampling effort is equal across all parcels, the identifier of those parces	NA	A
tree_id	chr	The identifier for a tree within a plot, ideally a number increasing from 1 to the number of trees in a plot, but this is not strict	NA	45
stem_id	chr	The identifier for a stem within a tree, ideally a number increasing from 1 to the number of stems in a tree, but this is not strict	NA	6
tag_id	chr	The code of the tag attached to the tree or stem in the field. If tags were not used, leave this column blank	NA	BP094
measurement_date	chr	The year the measurement was taken.	NA	1993
species_orig_binom	chr	The binomial name of the tree species, as provided by the data originator	NA	Burkea africana
species_orig_local	chr	The local name of the tree species/morphospecies, as provided by the data originator	NA	mumwei
genus	chr	The genus of the tree taken from 'species_orig_binom'	NA	Burkea
species	chr	The species epithet of the tree taken from 'species_orig_binom'	NA	africana
confer	chr	If a species is identified only as cf. (e.g. <i>Burkea cf. africana</i>), then the species epithet goes here, taken from 'species_orig_binom'	NA	africana
subspecies	chr	If a species is identified with a subsp. (e.g. <i>Burkea africana</i> subsp. <i>longifolia</i>), then the subspecies epithet goes here, taken from 'species_orig_binom'	NA	longifolia
variety	chr	If a species is identified with a variety (e.g. <i>Burkea africana</i> var. <i>thornii</i>), then the variety epithet goes here, taken from 'species_orig_binom'	NA	thornii
notes_stem	chr	Miscellaneous notes related to the tree or stem	NA	Leaning around 45 degrees and with a beehive in the fluted trunk
diam	num	Stem diameter measurement in cm	cm	6.4
pom	num	The height at which diam was taken, in metres to one decimal place (nearest 10 cm)	m	1.3
x_grid	num	The x coordinate of the stem within the plot if a grid method was used. In metres to nearest 10 cm	m	54.2

Table 9: Column names for stems table. (*continued*)

Column	Class	Description	Units	Example
y_grid	num	The y coordinate of the stem within the plot if a grid method was used. In metres to nearest 10 cm	m	35.6
latitude	num	The latitude coordinate of the stem within the plot if a GPS method was used. WGS84 EPSG4326	decimal degrees	-50.425622
longitude	num	The longitude coordinate of the stem within the plot if a GPS method was used. WGS84 EPSG4326	decimal degrees	30.446453
angle	num	If 'plot_shape' == "circle", and angular stem coordinates were used, the angle (0 = North) direction of the stem from the plot centre, in degrees (0-360)	degrees	43.2
distance	num	If 'plot_shape' == "circle", and angular stem coordinates were used, the distance of the stem from the plot centre. In metres to the nearest 10 cm	m	3.5
height	num	The height of the tree, in metres to one decimal place (nearest 10 cm)	m	3.5
bole_height	num	The height from the ground to the base of the lowest branch also called 'clear length'	m	6.3
crown_x	num	If two crown diameters are taken perpendicular to one another, the first of these measurements. If one crown diameter was taken, that crown diameter, in metres to one decimal place (nearest 10 cm).	m	5.2
crown_y	num	If two crown diameters are taken perpendicular to one another, the second of these measurements. If one crown diameter was taken, this remains blank, in metres to one decimal place (nearest 10 cm).	m	3.5
recruit	logi	Is the stem a new recruit since the last census? If this is the first census, leave blank.	NA	TRUE
on_termite	logi	Is the stem located on a termite mound?	NA	FALSE
lightning	logi	Has the stem been struck by lightning?	NA	TRUE
canopy_climbers	logi	Does the stem have other climbing plants in its canopy?	NA	FALSE
stem_stranglers	logi	Does the stem have strangling plants on it?	NA	TRUE
stump	logi	Is the stem a stump? I.e. has it been cut or broken lower than the POM?	NA	TRUE
liana	logi	Is the stem a liana, i.e. a woody stem that is not self-supporting and uses other trees to grow towards the canopy	NA	FALSE
alive	fct	Is the stem alive or dead. Possible values: "a" (alive), "d" (dead)	NA	A

Table 9: Column names for stems table. (*continued*)

Column	Class	Description	Units	Example
stem_status	fct	What is the growth status of the stem. Possible values are: "r" if the stem is topkilled and resprouting, "t" if the stem is topkilled with no sign of resprouting, "d" if the stem is completely dead (i.e. base appears dead), NA if the stem is alive with no abnormal growth status.	NA	R
standing_fallen	fct	Is the stem standing, or fallen? Possible values are "s" if the stem is standing, "f" if the stem is fallen at a clearly unnatural angle.	NA	F
broken_per_remain	num	If the stem is broken, approximately how much is remaining? As a decimal, e.g. 60% of stem remaining = 0.6, to one decimal place (nearest 10%)	proportion	0.2
broken_height	num	If the stem is broken, at what height is it broken. In metres to one decimal place (nearest 10 cm)	m	4.5
stem_mode	fct	What position is the stem in? Possible values are: "u" if uprooted, "p" if snapped, "s" if standing, "v" if vanished (location of tree found), "q" if unknown, "x" if location of tree not found due to problems, poor maps, etc.	NA	P
damage_cause	fct	If the stem is damaged, what caused it? Possible values are: "n" if neighbouring tree, "e" if elephants, "f" if fire, "h" if humans (.e.g. cut, ringbarked, broken or beehive), "l" if lightning, "m" if termites, "w" if wind, "q" if unknown	NA	N
damage_cause_human	fct	If the stem is damaged, and damage_cause = "h". What was the specific method of damage by humans. Possible values are: "chainsaw" if cut by a chainsaw, "axe" if cut by an axe, "machete" if cut by machete, "ringbark" if ringbarked, "beehive" if the bark was removed to access a beehive, "other".	NA	chainsaw
cambium	int	The proportion of trunk surface area with exposed vascular cambium, between the ground and 2 m.	proportion	0.22
bracket_fungi	logi	Does the stem have bracket fungi?	NA	TRUE
decay	num	What level of decay is the stem? Scale runs from 1 (no decay) to 5 (very decayed). Follow the "Decay class" table in the SEOSAW protocol. To 0.5 of a decay level (e.g. 1.5, 1, 3.5)	NA	4.5

Table 9: Column names for stems table. (*continued*)

Column	Class	Description	Units	Example
diam_method	fct	What method was used to derive the stem diameter measurement? Possible values are: "dbh_tape" if a tape measure with a diameter conversion scale was used, "tape" if a normal tape measure was used and the circumference converted to diameter, "caliper_1" if a single caliper measurement was used, "caliper_2" if two caliper measurements at 90 degrees with the mean taken, "other".	NA	dbh_tape
height_method	fct	What method was used to derive the height measurement? Possible values are: "eye" if estimated by eye, "rangefinder" if estimated using laser or ultrasonic apparatus with an electronic tilt sensor e.g. Nikon Forestry Pro, "clinometer" if estimated using a mechanical Clinometer, "hyposometer" if estimated using a Hagl�f Vertex Hypsometer, "direct" if estimated by climbing the tree or adjacent tower and measuring the height with tape or stick, "other"	NA	eye
voucher_id	chr	If a herbarium voucher has been collected, what is the code to identify it? 1, Wedza_01, MHR-04b are all valid. No spaces please.	NA	KEW_FK_42
fpc	num	Fractional Probability of Inclusion. If plots are nested with varying sampling effort within plots for stems of different size, the probability of a stem this size occurring in the whole plot.	proportion	0.35
measurement_id	chr	A standardised method of identifying measurements. A composite of 'plot_id' 'tree_id' 'stem_id' 'measurement_date'	NA	ABG_1_40_3_1990
crown_area	num	Area of the tree crown, measured as an ellipse using either one or two crown diameter measurements, if present. In m^2	m^2	4.5
agb	num	Aboveground biomass of stem, in tonnes of dry matter	t DM	0.093
ba	num	Basal area of stem, in m^2	m^2	0.46
diam_adj	num	Adjusted stem diameter at 1.3 m height accounting for tree taper	cm	5.3
species_name_clean	chr	Cleaned species name after taxize checks and synonymy corrections	NA	Burkea africana
genus_clean	chr	Cleaned genus name, taken from 'species_name_clean'	NA	Burkea
species_clean	chr	Cleaned species epithet, taken from 'species_name_clean'	NA	africana
subspecies_clean	chr	Cleaned subspecies epithet, taken from 'species_name_clean'	NA	africana
confer_clean	chr	Cleaned cf. part of species epithet, taken from 'species_name_clean'	NA	polyantha
variety_clean	chr	Cleaned var. part of species epithet, taken from 'species_name_clean'	NA	micrantha
family_gnr	chr	Matched family name using taxize::gnr_resolve()	NA	Fabaceae
name_gnr	chr	Name matched during the GNR resolve	NA	Burkea africana
taxon_rank_gnr	chr	All rank names of classification from GNR resolve, separated by " "	NA	superkingdom kingdom phylum subphylum

Table 9: Column names for stems table. (*continued*)

Column	Class	Description	Units	Example
taxon_gnr	chr	All ranks of classification from GNR resolve, separated by " "	NA	Eukaryota Viridiplantae Streptophyta St.
base_rank_gnr	chr	The lowest rank identified in GNR resolve	NA	species
name_cjb	chr	Full species name with authority matched with the CJB African Plants database	NA	Burkea africana Host.
url_cjb	chr	URL of species record matched with the CJB African Plants database	NA	https://www.ville-ge.ch/...
proj_SA_cjb	logi	Is the species part of the CJB Southern African Area?	NA	TRUE
proj_TA_cjb	logi	Is the species part of the CJB Tropical African Area?	NA	FALSE
proj_NA_cjb	logi	Is the species part of the CJB Northern African Area?	NA	TRUE
proj_MA_cjb	logi	Is the species part of the CJB Madagascar Area?	NA	FALSE

Table 10: Column names for plot spatial data table.

Column	Class	Description	Resolution	Citation	Units
plot_id	chr	A standardised method of identifying plots. For example, the first plot in a set of plots installed in Angola (A), in Bicular National Park (B), by John Godlee (G), would be referred to as ABG_1. The three letter code should be conserved within a dataset, i.e. a group of plots owned by a particular researcher within a particular locale.	NA	NA	NA
bio1	num	WorldClim - Mean Annual Temperature	30s	Fick et al. 2017	degC
bio2	num	WorldClim - Mean Diurnal Range (mean monthly max. temp. - min. temp.)	30s	Fick et al. 2017	degC
bio3	num	WorldClim - Isothermality (bio2 / bio7)	30s	Fick et al. 2017	degC
bio4	num	WorldClim - Temperature Seasonality (standard deviation of bio1)	30s	Fick et al. 2017	degC
bio5	num	WorldClim - Max Temperature of Warmest Month	30s	Fick et al. 2017	degC
bio6	num	WorldClim - Min Temperature of Coldest Month	30s	Fick et al. 2017	degC
bio7	num	WorldClim - Temperature Annual Range (mean of annual bio5 - bio6)	30s	Fick et al. 2017	degC
bio8	num	WorldClim - Mean Temperature of Wettest Quarter	30s	Fick et al. 2017	degC
bio9	num	WorldClim - Mean Temperature of Driest Quarter	30s	Fick et al. 2017	degC
bio10	num	WorldClim - Mean Temperature of Warmest Quarter	30s	Fick et al. 2017	degC
bio11	num	WorldClim - Mean Temperature of Coldest Quarter	30s	Fick et al. 2017	degC

Table 10: Column names for plot spatial data table. (*continued*)

Column	Class	Description	Resolution	Citation	Units
bio12	num	WorldClim - Mean Annual Precipitation	30s	Fick et al. 2017	mm y ⁻¹
bio13	num	WorldClim - Precipitation of Wettest Month	30s	Fick et al. 2017	mm
bio14	num	WorldClim - Precipitation of Driest Month	30s	Fick et al. 2017	mm
bio15	num	WorldClim - Precipitation Seasonality (Coefficient of Variation of bio12)	30s	Fick et al. 2017	mm
bio16	num	WorldClim - Precipitation of Wettest Quarter	30s	Fick et al. 2017	mm
bio17	num	WorldClim - Precipitation of Driest Quarter	30s	Fick et al. 2017	mm
bio18	num	WorldClim - Precipitation of Warmest Quarter	30s	Fick et al. 2017	mm
bio19	num	WorldClim - Precipitation of Coldest Quarter	30s	Fick et al. 2017	mm
phen_avg_vi	num	Mean annual EVI (Enhanced Vegetation Index)	5600 m	Didan and Barretto 2016	NA
phen_cum_vi	num	Cumulative annual EVI (Enhanced Vegetation Index)	5600 m	Didan and Barretto 2016	NA
phen_day_peak_1	num	Mean day of year of peak EVI of growing season 1	5600 m	Didan and Barretto 2016	NA
phen_day_peak_2	num	Mean day of year of peak EVI of growing season 2	5600 m	Didan and Barretto 2016	NA
phen_day_peak_3	num	Mean day of year of peak EVI of growing season 3	5600 m	Didan and Barretto 2016	NA
herbiv_total	num	Mean total herbivore biomass	0.5deg	Hempson et al. 2017	kg km ⁻²
herbiv_small_nonsocial_browser	num	Mean total Small non-social browsers biomass	0.5deg	Hempson et al. 2017	kg km ⁻²
herbiv_medium_sized_social	num	Mean total Medium-sized social mixed herbivore biomass	0.5deg	Hempson et al. 2017	kg km ⁻²
herbiv_large_browser	num	Mean total Large browser biomass	0.5deg	Hempson et al. 2017	kg km ⁻²
herbiv_water_dependent	num	Mean total Water-dependent grazers biomass	0.5deg	Hempson et al. 2017	kg km ⁻²

Table 10: Column names for plot spatial data table. (*continued*)

Column	Class	Description	Resolution	Citation	Units
herbiv_non_ruminant	num	Mean total Non-ruminants biomass	0.5deg	Hempson et al. 2017	kg km ⁻²
herbiv_elephant	num	Mean elephant biomass	0.5deg	Hempson et al. 2017	kg km ⁻²
elevation	num	The elevation of the plot. Measured in metres above International Mean sea level to 1 decimal place. From SRTM	3s	Jarvis et al. 2008	m asl
tpi	num	Topographic Position Index - difference between cell elevation and mean value of 8 surrounding cells	1000 m	Amatulli et al. 2018	m asl
tri	num	Terrain Ruggedness Index - mean of absolute difference between cell elevation and value of 8 surrounding cells	1000 m	Amatulli et al. 2018	m asl
roughness	num	Topographic Roughness Index - difference between maximum and minimum cell elevation and its 8 surrounding cells	1000 m	Amatulli et al. 2018	m asl
slope	num	Slope	1000 m	Amatulli et al. 2018	degrees
admin_region	chr	Administrative area at level 1 according to rnatuarearth data	NA	South 2017	NA
soil_bulk_density	num	Bulk density of soil (fine earth)	250 m	Hengl et al. 2017	kg m ⁻³
soil_cation_ex_cap	num	Cation Exchange Capacity of soil	250 m	Hengl et al. 2017	cmolc kg ⁻¹
soil_clay	num	Clay content of soil (0-2 um) mass fraction	250 m	Hengl et al. 2017	proportion
soil_sand	num	Sand content of soil (50-2000 um) mass fraction	250 m	Hengl et al. 2017	proportion
soil_silt	num	Silt content of soil (2-50 um) mass fraction	250 m	Hengl et al. 2017	proportion
soil_coarse_frag	num	Coarse fragments volumetric in soil	250 m	Hengl et al. 2017	proportion
soil_ph	num	Soil pH x 10 in H2O	250 m	Hengl et al. 2017	NA
soil_org_c	num	Soil organic carbon stock	250 m	Hengl et al. 2017	Mg ha ⁻¹
soil_org_c_densit	num	Soil organic carbon density	250 m	Hengl et al. 2017	kg m ⁻³
soil_nitrogen	num	Soil nitrogen content	250 m	Hengl et al. 2017	g kg ⁻¹

Table 10: Column names for plot spatial data table. (*continued*)

Column	Class	Description	Resolution	Citation	Units
teow_eco	chr	Dinerstein et al. 2017 - Terrestrial Ecoregion	NA	Dinerstein et al. 2017	NA
teow_nnh	chr	Dinerstein et al. 2017 - Nature Needs Half classification	NA	Dinerstein et al. 2017	NA
teow_realm	chr	Dinerstein et al. 2017 - Terrestrial Realm	NA	Dinerstein et al. 2017	NA
teow_biome	chr	Dinerstein et al. 2017 - Terrestrial Biome	NA	Dinerstein et al. 2017	NA
whites_veg	chr	Whites vegetation map of Africa (1983)	NA	Dinerstein et al. 2017	NA
frost_days	num	Mean annual number of frosty days (1990-2019)	0.5deg	Harris et al. 2020	NA
fire_count	int	MODIS burned area - Number of fires between 2001 and 2018	1000 m	Giglio et al. 2015	n
fire_intensity	num	MODIS active fires - mean fire intensity	1000 m	Giglio et al. 2016	NA
forest_height	num	Woody canopy height, estimated from GLAD GEDI Global Forest Height	30 m	Potapov et al. 2020	m
travel_time_city	num	Malaria Atlas Project, travel time to nearest city (>50,000 people)	30s	Nelson et al. 2019	min
travel_time_settlement	num	Malaria Atlas Project, travel time to nearest settlement (>5000 people)	30s	Nelson et al. 2019	min
distance_road	num	GRIP4 distance to nearest road of any size	NA	Meijer et al. 2018	m
type_nearest_road	chr	GRIP4 type of the nearest road	NA	Meijer et al. 2018	NA
pop_density	num	Worldpop Population density - people per km2	1000 m	Linard et al. 2012	n km ²

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